NeoSignal

AI Intelligence Daily · Multi-Source · Cross-Verified

07 July 2026

250 verified · 512 raw · 5 sources

250	4	246	0
TOTAL	VERIFIED	CONFIRMED	EMERGING

Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

VERIFIED	**0.80 - 1.00**	3+ independent sources including media
CONFIRMED	**0.50 - 0.79**	2 sources or 1 high-quality media
EMERGING	**0.25 - 0.49**	Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	237	Media
HackerNews	7	Community
TechCrunch AI	7	Media
MIT Tech Review	2	Media
The Verge AI	1	Media

VERIFIED INTELLIGENCE — 4 stories

VERI

Show HN: Dedtxt is a quiet, cross-platform plain-text editor

HackerNews +1 source 90% HN 4

arXiv:2604.10311v2 Announce Type: replace Abstract: Artificial Intelligence (AI) models, encompassing both traditional machine learning (ML) and more advanced approaches such as deep learning and large language models (LLMs), play a central role in modern applications. AI model...

https://dedtxt.app/

VERI

Show HN: Built multi-hop passenger transfer routing for a browser train SIM

HackerNews +1 source 90% HN 2

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. LLMs are stuck in a groupthink groove. This startup is trying to get them out. Open up your chatbot of choice—Claude, ChatGPT, Gemini—and...

https://overlandgame.netlify.app/

VERI

dOPSD: On-Policy Self-Distillation for Diffusion Language Models

ArXiv CS.AI +1 source 80%

arXiv:2607.04428v1 Announce Type: cross Abstract: Diffusion large language models (dLLMs) generate text by iteratively denoising a masked sequence, offering a parallel alternative to autoregressive models, but eliciting strong reasoning through post-training remains difficult:...

https://arxiv.org/abs/2607.04428

VERI

Some of the nation's rich are letting AI teach their kids

The Verge AI +1 source 80%

Most Americans don't trust AI. It's proven that it doesn't know what safe toppings for pizza are. People don't even want to listen to AI music. But none of that matters for some of America's wealthy, who are turning to AI to teach their kids instead of traditional schools....

https://www.theverge.com/ai-artificial-intelligence/961505/wealthy-ai-schools-alpha-for...

CONFIRMED SIGNALS — 246 stories

CONF

Show HN: Scan your AI agents for dangerous capabilities

HackerNews 50% HN 41

At the AI Engineering World's Fair a big part of the conversation was to nail the self improvement loop. Our take on this is to record state of the agent execution with a durable runtime, then allow users to replay from state checkpoints and run 'what-if' experiments. It's OSS...

https://github.com/makerchecker/MakerChecker

CONF

Show HN: LLM Thought Visualization

HackerNews 50% HN 18

No summary — see link for full article.

https://github.com/ninjahawk/Subtext

CONF

Show HN: Paint the Earth on a live, interactive globe (collaborative art.)

HackerNews 50% HN 18

Earth.tattoo divides the earth into 510 million 16x16 pixel "tiles" that you can own and paint as you like. You can claim one free tile per hour.

https://earth.tattoo

CONF

Show HN: Otari: your open-source LLM control plane

HackerNews 50% HN 16

i, like you, probably burn tokens like water and i don't trust the /status or /usage calls b/c, for whatever reason, they either fail to load in the CLI). so, i'm constantly refreshing the direct links to Claude / Codex dashboard, manually. this, of course, is not reasonable....

https://github.com/mozilla-ai/otari

CONF

Show HN: Grinta –local-first AI coding agent that passed 106-minute stress test

HackerNews 50% HN 2

No summary — see link for full article.

https://github.com/josephsenior/Grinta-Coding-Agent

0 CONF

The ‘first’ AI-run ransomware attack still needed a human

TechCrunch AI 50%

An AI agent carried out the technical execution of a real-world ransomware attack for the first known time, but new details show a human still chose the victim, set up the infrastructure, and supplied stolen credentials — meaning it wasn't quite the fully autonomous cybercrime...

https://techcrunch.com/2026/07/06/the-first-ai-run-ransomware-attack-still-needed-a-human/

1 CONF

US investors will soon get access to SK Hynix, another memory maker riding the AI boom

TechCrunch AI 50%

SK Hynix is experiencing a boom credited to AI. It will ride that to a multibillion-dollar U.S. IPO, expected to take place on Friday.

https://techcrunch.com/2026/07/06/us-investors-will-soon-get-access-to-sk-hynix-another...

2 CONF

Every major tech layoff in 2026 that has name-checked AI

TechCrunch AI 50%

A running look — in reverse chronological order — at the bigger tech companies that have announced significant layoffs this year with AI as a stated factor.

https://techcrunch.com/2026/07/06/the-running-list-major-tech-layoffs-in-2026-where-emp...

3 CONF

If you use Google, you’re training its AI. Here’s how to opt out.

TechCrunch AI 50%

Consider this a belated PSA: A recent change to Google’s privacy settings is allowing the company to store more of your data, including media such as “images, files, and audio and video recordings,” to improve its AI models.

https://techcrunch.com/2026/07/06/if-you-use-google-youre-training-its-ai-heres-how-to-...

4 CONF

Reddit is using LLMs to solve a problem LLMs largely created

TechCrunch AI 50%

In the AI era, platforms have no choice but to fight fire with fire to cull spam.

https://techcrunch.com/2026/07/06/reddit-is-using-llms-to-solve-a-problem-llms-largely-...

5 CONF

Station F ramps up as a launchpad for Europe's hottest AI startups

TechCrunch AI 50%

Station F, a Paris-based startup hub founded by French billionaire Xavier Niel, is gearing up for a new edition of its F/ai accelerator program in a bid to strengthen its positioning as a stepping stone for promising AI startups.

https://techcrunch.com/2026/07/06/station-f-ramps-up-as-a-launchpad-for-europes-hottest...

6 CONF

Your family's $300 stake in OpenAI

MIT Tech Review 50%

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. OpenAI CEO Sam Altman's oft-discussed promise that Americans will share in the wealth AI creates was in the news again last week. On...

https://www.technologyreview.com/2026/07/06/1140176/your-familys-300-stake-in-openai/

7 CONF

Graph Neural Networks are Heuristics

ArXiv CS.AI 50%

arXiv:2601.13465v4 Announce Type: replace Abstract: Graph neural networks are usually treated as auxiliaries for combinatorial optimization: they imitate algorithms, guide search, or supply scores to classical procedures. We show that this auxiliary role is not intrinsic. A GNN...

https://arxiv.org/abs/2601.13465

8 CONF

ASK in the Dark: Uncertainty-Gated LLM Assistance under Partial Observability

ArXiv CS.AI 50%

arXiv:2607.02686v1 Announce Type: new Abstract: Reinforcement learning agents operating under partial observability must act on incomplete information, making them natural candidates for guidance from small language models (SLMs) that carry broad reasoning priors. Yet...

https://arxiv.org/abs/2607.02686

9 CONF

Mask-based Predictive Representations for Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2607.04153v1 Announce Type: cross Abstract: Vision-based deep reinforcement learning involves dealing with high-dimensional inputs of image information. It is crucial to abstract effective states from high-dimensional image inputs and limited samples for sample-efficient...

https://arxiv.org/abs/2607.04153

0 CONF

DualView: Preventing Indirect Prompt Injection in Personal AI Agents

ArXiv CS.AI 50%

arXiv:2607.03821v1 Announce Type: cross Abstract: Personal AI agents that run on the user's local machine, such as OpenClaw, automate daily tasks including web search, email, and file management. Their access to computer resources, including the network, file system, and shell,...

https://arxiv.org/abs/2607.03821

1 CONF

ELBO-T2IAlign: A Generic ELBO-Based Method for Calibrating Pixel-level Text-Image Alignment in Diffusion Models

ArXiv CS.AI 50%

arXiv:2506.09740v2 Announce Type: replace-cross Abstract: Diffusion models excel at image generation. Recent studies have shown that these models not only generate high-quality images but also encode text-image alignment information through attention maps or loss functions. This...

https://arxiv.org/abs/2506.09740

2 **CONF**

CRRL: A Causality-Based Reinforcement Learning Framework for Autonomous System Recovery

ArXiv CS.AI 50%

arXiv:2607.03177v1 Announce Type: cross Abstract: Traditional reinforcement learning (RL) for recovery in autonomous systems lacks causal understanding and generalizes poorly to novel failure scenarios. RL policies often stall in failure states, spending up to 70% of an episode...

https://arxiv.org/abs/2607.03177

3 **CONF**

Efficient bias mitigation in T2I diffusion models using Concept Graphs

ArXiv CS.AI 50%

arXiv:2607.03397v1 Announce Type: new Abstract: Text-to-Image diffusion models often propagate harmful bias inherited from the training data. Existing bias mitigation techniques typically intervene only at the text encoder or provide inference-time guidance, often leading to...

https://arxiv.org/abs/2607.03397

4 **CONF**

Embodied Operators and Benchmarking: Toward Reusable and Deployable Embodied Intelligence Systems

ArXiv CS.AI 50%

arXiv:2607.03283v1 Announce Type: new Abstract: Embodied intelligence systems require not only end-to-end policy models, but also reusable functional modules that transform multimodal observations, robot states, human demonstrations, and task contexts into structured...

https://arxiv.org/abs/2607.03283

5 **CONF**

Reflective Dialogue or Prompt Refinement? Effects of Tutor Scaffolding on Students' Independent LLM Use for Programming

ArXiv CS.AI 50%

arXiv:2607.03303v1 Announce Type: new Abstract: While Large Language Models (LLMs) can provide personalized support in learning, several studies have raised concerns regarding their use in education. Importantly, learning depends on how students engage with LLMs. This study...

https://arxiv.org/abs/2607.03303

6 **CONF**

When Aggregate Alignment Misleads: Auditing Policy Repair Without Per-State Expert Actions

ArXiv CS.AI 50%

arXiv:2607.03386v1 Announce Type: new Abstract: Agentic AI systems are increasingly used to edit, refine, and repair decision policies, but evaluating these edits is difficult when per-state expert action labels are unavailable. We study this problem in a hotel-pricing simulator...

https://arxiv.org/abs/2607.03386

7 **CONF**

Demonstrating Generalization Failures via Mixtures of Conditional Policies

ArXiv CS.AI 50%

arXiv:2607.03478v1 Announce Type: new Abstract: Post-training of frontier language models is conducted on curated task suites, and inevitably leaves a distribution shift between training and deployment environments. This exposes developers to generalization failures, which are...

https://arxiv.org/abs/2607.03478

8 **CONF**

How to Avoid Debate: Scalable AI Safety via Doubly-Efficient Interactive Proofs

ArXiv CS.AI **50%**

arXiv:2607.03561v1 Announce Type: new Abstract: As AI models continue to develop powerful capabilities, it becomes critical that we are able to verify that their output is aligned with our intentions. A recent line of work focuses on verification via debate, a model of...

https://arxiv.org/abs/2607.03561

9 **CONF**

The Changing Role of Symbolic Methods in Artificial Intelligence

ArXiv CS.AI **50%**

arXiv:2607.05168v1 Announce Type: new Abstract: Why do intelligent systems need to perform explicit symbolic reasoning? Computer science has traditionally regarded symbolic reasoning as a defining component of intelligence. Yet the remarkable success of modern foundation models...

https://arxiv.org/abs/2607.05168

0 **CONF**

Agent Reinforcement Learning via Pivotal-Aware Self-Feedback Retry

ArXiv CS.AI **50%**

arXiv:2607.03702v1 Announce Type: new Abstract: Large language model (LLM) agents have shown strong decision-making capabilities in long-horizon interactive tasks, yet they still struggle to effectively leverage failed trajectories: full retries incur high interaction costs,...

https://arxiv.org/abs/2607.03702

1 **CONF**

Evaluating LLM-Based Regression Test Generation

ArXiv CS.AI **50%**

arXiv:2501.11086v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) have shown tremendous promise in automated software engineering. In this paper, we investigate LLMs for just-in-time regression test generation for programs, like parsers, interpreters, or...

https://arxiv.org/abs/2501.11086

2 **CONF**

VideoSearcher: Empowering Video Deep Research with Multi-Tool Agentic Reasoning via Reinforcement Learning

ArXiv CS.AI **50%**

arXiv:2607.02927v1 Announce Type: cross Abstract: Video understanding is moving beyond closed-context perception toward open-world evidence exploration, a paradigm formalized as Video Deep Research (VDR). However, existing multimodal search agents primarily target static images,...

https://arxiv.org/abs/2607.02927

3 **CONF**

Explainable AI for Screening Abuse-Related Trauma in Bangladeshi Children: A Training-Free Multimodal Framework Evaluated on Noise-Aware Synthetic Data

ArXiv CS.AI **50%**

arXiv:2607.04010v1 Announce Type: new Abstract: Bangladesh has an estimated 1.17 mental-health professionals per 100,000 population and only six child psychiatrists nationwide. No Bengali-language, culturally adapted tool exists for early screening of abuse-related psychological...

https://arxiv.org/abs/2607.04010

4 CONF

The Rise of Large Language Models and the Direction and Impact of US Federal Research Funding

ArXiv CS.AI 50%

arXiv:2601.15485v3 Announce Type: replace-cross Abstract: Federal research funding shapes the direction, diversity, and impact of the US scientific enterprise. Large language models (LLMs) are rapidly diffusing into scientific practice, holding substantial promise while raising...

https://arxiv.org/abs/2601.15485

5 CONF

Don't Blame the Large Language Model: How Scaffolding Evolution Shapes Coding Agent Quality

ArXiv CS.AI 50%

arXiv:2607.03691v1 Announce Type: cross Abstract: Coding agents, autonomous systems that use large language models (LLMs) to resolve software engineering tasks, rely on agentic scaffolding: a middleware layer in between a developer and a large language model that orchestrates...

https://arxiv.org/abs/2607.03691

6 CONF

Progress- and Reliability-Oriented Group Policy Optimization for Agentic Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2607.04242v1 Announce Type: new Abstract: Group-based reinforcement learning (RL) has become an effective paradigm for improving large language model agents on long-horizon interactive tasks. To obtain finer-grained policy updates than trajectory-level optimization, recent...

https://arxiv.org/abs/2607.04242

7 CONF

Agentic SABRE: An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection

ArXiv CS.AI 50%

arXiv:2607.04292v1 Announce Type: new Abstract: Ransomware has evolved into a complex, adaptive, and fast-moving adversary category in which static signatures and monolithic classifiers fail to generalise under concept drift, evasion, and behavioural polymorphism. In this paper,...

https://arxiv.org/abs/2607.04292

8 CONF

PPE-Bench: A Benchmark for Evaluating MLLM Unlearning under Private-Public Entanglement

ArXiv CS.AI 50%

arXiv:2607.02897v1 Announce Type: cross Abstract: Multimodal Large Language Models (MLLMs) have shown strong capabilities, but they may memorize private information from web data, raising privacy concerns. Machine unlearning offers a way to remove such private knowledge without...

https://arxiv.org/abs/2607.02897

9 CONF

Server-side Anti-cheat in FPS games for Aimbot detection using Deep learning and Machine learning

ArXiv CS.AI 50%

arXiv:2607.04336v1 Announce Type: new Abstract: Modern video games are becoming more complex day by day. Most of these modern games are multiplayer first-person shooter (FPS) games. The rising popularity of FPS games emphasizes the need to combat cheating for fair and enjoyable...

https://arxiv.org/abs/2607.04336

0 **CONF**

Nemotron-Labs-3-Puzzle-75B-A9B: Compressing Hybrid MoE LLMs

ArXiv CS.AI 50%

arXiv:2607.04371v1 Announce Type: new Abstract: We present Nemotron-Labs-3-Puzzle-75B-A9B, a compressed variant of Nemotron-3-Super optimized for interactive deployment. We designed the model to maximize server throughput under high user throughput constraints. In interactive...

https://arxiv.org/abs/2607.04371

1 **CONF**

Decentralized Aggregation of LLM Predictions via Wagering Mechanisms

ArXiv CS.AI 50%

arXiv:2607.04389v1 Announce Type: new Abstract: It is increasingly common to aggregate predictions from multiple LLMs, each with domain expertise or access to private tools and data, to improve collective prediction performance. In decentralized settings, aggregation weights...

https://arxiv.org/abs/2607.04389

2 **CONF**

LLM-as-a-Tutor: Policy-Aware Prompt Adaptation for Non-Verifiable RL

ArXiv CS.AI 50%

arXiv:2607.04412v1 Announce Type: new Abstract: Reinforcement learning (RL) for non-verifiable instruction following increasingly relies on LLM judges with prompt-specific rubrics as reward signals. While recent methods adapt these rubrics to the evolving policy during training,...

https://arxiv.org/abs/2607.04412

3 **CONF**

Agent Step Value: State-Transition Measurement with State-Grounded LLM Evaluators

ArXiv CS.AI 50%

arXiv:2607.04419v1 Announce Type: new Abstract: Most agent evaluations collapse a multi-step trace into a final answer, a success flag, or a trajectory-level score. These aggregates obscure the diagnostic question developers need most: which action changed the state in a useful...

https://arxiv.org/abs/2607.04419

4 **CONF**

OmniFocus: Query-Guided Modality-Balanced Token Compression for Omni-Modal Large Language Models

ArXiv CS.AI 50%

arXiv:2607.03050v1 Announce Type: cross Abstract: Omni modal large language models (OmniLLMs) have attracted wide attention for their ability to jointly process audio and video, but they generate large token sequences under audio-visual inputs, leading to substantial inference...

https://arxiv.org/abs/2607.03050

5 **CONF**

Detecting Answer-Driven Reasoning in LLM-Based Educational Tutors via Truncated Chain-of-Thought Auditing

ArXiv CS.AI 50%

arXiv:2607.04572v1 Announce Type: new Abstract: Large language model (LLM) tutors often produce fluent step-by-step explanations, but a correct and pedagogically formatted response does not guarantee that the answer was derived from the student-facing problem. In realistic...

https://arxiv.org/abs/2607.04572

6 CONF

MRMS: A Multi-Resolution Memory Substrate for Long-Lived AI Agents

ArXiv CS.AI 50%

arXiv:2607.04617v1 Announce Type: new Abstract: Long-lived AI agents require continuity across interactions, but continuity cannot be obtained by simply extending the prompt window. An agent must preserve useful prior experience, retrieve it selectively, distinguish personal...

https://arxiv.org/abs/2607.04617

7 CONF

Integrated Altruistic and Fairness Preference Induces Advanced Mutual Cooperation in Sequential Social Dilemmas

ArXiv CS.AI 50%

arXiv:2607.04710v1 Announce Type: new Abstract: Inducing cooperation among distributed agents is still a difficult problem in the field of multi-agent reinforcement learning (MARL), particularly in social dilemma situations. There, individual interests are misaligned with the...

https://arxiv.org/abs/2607.04710

8 CONF

AgentGym2: Benchmarking Large Language Model Agents in De-Idealized Real-World Environments

ArXiv CS.AI 50%

arXiv:2607.05174v1 Announce Type: new Abstract: Language agents, i.e., LLM agents, progress rapidly and are increasingly deployed in production environments. This trend underscores the urgent need for rigorous and realistic evaluations. However, most existing benchmarks evaluate...

https://arxiv.org/abs/2607.05174

9 CONF

A Unified Algebraic Framework for Classification Performance Evaluation

ArXiv CS.AI 50%

arXiv:2607.04028v1 Announce Type: cross Abstract: We propose a unified algebraic framework for classification performance evaluation that encompasses binary, multiclass, multilabel, ordinal, hierarchical, cost-sensitive, and soft-label settings within a single formalism. The...

https://arxiv.org/abs/2607.04028

0 CONF

ClassicLogic: A Knowledge-Driven Benchmark of Classic Puzzle Games for Evaluating Compositional Generalization

ArXiv CS.AI 50%

arXiv:2607.05185v1 Announce Type: new Abstract: Compositional generalization, the ability to understand and produce novel combinations of known components, remains a fundamental challenge for modern artificial intelligence. While few benchmarks exist, many focus on linguistic...

https://arxiv.org/abs/2607.05185

1 CONF

Reason, Reward, Refine: Step-Level Errors Corrections with Structured Feedback for Physics Reasoning in Small Language Models

ArXiv CS.AI 50%

arXiv:2607.05199v1 Announce Type: new Abstract: Physics reasoning fails structurally in small language models: an error at any step propagates forward, corrupting every inference that follows. Limited domain knowledge, hallucination under multi-step derivation, and...

https://arxiv.org/abs/2607.05199

2 CONF

EvoAgentBench: Benchmarking Agent Self-Evolution via Ability Transfer

ArXiv CS.AI 50%

arXiv:2607.05202v1 Announce Type: new Abstract: Agent self-evolution in long-horizon LLM systems is largely procedural: useful experience is not merely stored information, but reusable procedures for searching, debugging, and verification. Yet current evaluations do not isolate...

https://arxiv.org/abs/2607.05202

3 CONF

MetaSkill-Evolve: Recursive Self-Improvement of LLM Agents via Two-Timescale Meta-Skill Evolution

ArXiv CS.AI 50%

arXiv:2607.05297v1 Announce Type: new Abstract: Recent LLM agents tackle increasingly long-horizon, open-ended tasks, and external skills, reusable procedural knowledge supplied to the agent, further extend this capability. However, a fixed, hand-authored skill is rarely...

https://arxiv.org/abs/2607.05297

4 CONF

LAW & ORDER: Adaptive Spatial Weighting for Medical Diffusion and Segmentation

ArXiv CS.AI 50%

arXiv:2603.04795v2 Announce Type: replace-cross Abstract: Medical image analysis depends on accurate segmentation and controllable synthesis, but both tasks face severe spatial imbalance: lesions occupy small regions against large backgrounds. We study adaptive spatial weighting...

https://arxiv.org/abs/2603.04795

5 CONF

LLM-as-a-Verifier: A General-Purpose Verification Framework

ArXiv CS.AI 50%

arXiv:2607.05391v1 Announce Type: new Abstract: Scaling pre-training, post-training, and test-time compute have become the central paradigms for improving the capabilities of LLMs. In this work, we identify verification, the ability to determine the correctness of a solution, as...

https://arxiv.org/abs/2607.05391

6 CONF

SiamixFormer: a fully-transformer Siamese network with temporal Fusion for accurate building detection and change detection in bi-temporal remote sensing images

ArXiv CS.AI 50%

arXiv:2208.00657v2 Announce Type: cross Abstract: Building detection and change detection using remote sensing images can help urban and rescue planning. Moreover, they can be used for building damage assessment after natural disasters. Currently, most of the existing models for...

https://arxiv.org/abs/2208.00657

7 CONF

PotatoGANs: Utilizing Generative Adversarial Networks, Instance Segmentation, and Explainable AI for Enhanced Potato Disease Identification and Classification

ArXiv CS.AI 50%

arXiv:2405.07332v2 Announce Type: cross Abstract: Numerous applications have resulted from the automation of agricultural disease segmentation using deep learning techniques. However, when applied to new conditions, these applications frequently face the difficulty of...

https://arxiv.org/abs/2405.07332

8 **CONF**

GLM-5 Serving Parameter Tuning for OpenClaw: Single-Deployment MaaS Inference Optimization for Long-Context Agent Workloads

ArXiv CS.AI 50%

arXiv:2607.02518v1 Announce Type: cross Abstract: OpenClaw requests are dominated by long, tool-augmented prefixes, including system prompts, conversation history, and tool outputs fed back into the context window. For this workload, with about 28k-30k input tokens and 500...

https://arxiv.org/abs/2607.02518

9 **CONF**

From Tensor Buffer to Distributed Memory Hierarchy: A Survey of KV Cache Management for LLM Serving

ArXiv CS.AI 50%

arXiv:2607.02574v1 Announce Type: cross Abstract: The key-value (KV) cache has become a first-order memory object in LLM serving rather than a temporary per-request tensor. This survey classifies more than thirty KV-management systems and frameworks using four axes: locality,...

https://arxiv.org/abs/2607.02574

0 **CONF**

The Hidden Water Geography of U.S. Hyperscale Data Centers in the AI Era

ArXiv CS.AI 50%

arXiv:2607.02531v1 Announce Type: cross Abstract: Water use by data centers is routinely reported as a single footprint, but water is consumed through two physically distinct pathways: at the site for cooling and in the power system that generates electricity. We mapped both...

https://arxiv.org/abs/2607.02531

1 **CONF**

Not Every Sync Is Safe: Calibrated DiLoCo Scheduling for Shared AI Infrastructure

ArXiv CS.AI 50%

arXiv:2607.02544v1 Announce Type: cross Abstract: DiLoCo-style training reduces communication by letting learner islands train locally before occasional outer synchronization, making it attractive for fragmented industrial AI fleets where training shares hardware with...

https://arxiv.org/abs/2607.02544

2 **CONF**

A Random Matrix Theory Perspective on the Consistency of Diffusion Models

ArXiv CS.AI 50%

arXiv:2602.02908v2 Announce Type: replace-cross Abstract: Diffusion models trained on different, non-overlapping subsets of a dataset often produce strikingly similar outputs when given the same noise seed. We trace this consistency to a simple linear effect: the shared Gaussian...

https://arxiv.org/abs/2602.02908

3 **CONF**

DOSE-I: A Multimodal Biosignal Dataset of Procedural Sedation for Endoscopy -- Technical Report

ArXiv CS.AI 50%

arXiv:2607.02570v1 Announce Type: cross Abstract: In this document, we describe characteristics and technical details of the multimodal biosignal dataset DOSE-I of procedural sedation for endoscopy published on zenodo. The DOSE-I dataset includes 78.5 hours of recording in 171...

https://arxiv.org/abs/2607.02570

4 **CONF**

Consistent but Miscalibrated: Evaluating LLM Limitations for Risk Communication in Natural Language

ArXiv CS.AI 50%

arXiv:2607.03882v1 Announce Type: cross Abstract: LLMs are increasingly deployed as post-hoc explainers of AI-generated outputs, yet it remains unclear whether they can reliably communicate probabilistic information in natural language. For this role to be viable, models must...

https://arxiv.org/abs/2607.03882

5 **CONF**

An automated method of identifying incorrectly labelled images based on the sequences of loss functions of deep learning networks

ArXiv CS.AI 50%

arXiv:2607.02594v1 Announce Type: cross Abstract: Deep learning is widely applied in medical image analysis, but up to 10% of manually labelled images may be incorrect, degrading model performance. This paper proposes an automated method to identify incorrectly labelled medical...

https://arxiv.org/abs/2607.02594

6 **CONF**

A Retrieval-Augmented Framework for Detecting and Resolving Pragmatic Ambiguities in Natural Language Requirements

ArXiv CS.AI 50%

arXiv:2607.04436v1 Announce Type: cross Abstract: Natural language requirements (NLRs) are essential for bridging communication gaps among diverse stakeholders in software development. However, the inherent ambiguity in NLRs can pose significant challenges. In particular, some...

https://arxiv.org/abs/2607.04436

7 **CONF**

Fusion: A Framework for Unified Sequential Token AdaptatIon in VisiOn TraNsformers

ArXiv CS.AI 50%

arXiv:2607.02612v1 Announce Type: cross Abstract: Vision Transformers achieve strong image classification accuracy but process all image regions with nearly the same computation, even when many regions are redundant or uninformative. Recent adaptive inference methods reduce this...

https://arxiv.org/abs/2607.02612

8 **CONF**

COMET: Combinatorial Optimization for Multiplex Editing Targets Via Constraint-Preserving QAOA

ArXiv CS.AI 50%

arXiv:2607.02622v1 Announce Type: cross Abstract: Multiplex CRISPR-Cas9 gene editing requires selecting one guide RNA per target gene subject to cross-gene interactions: a constrained combinatorial problem that can be formulated as a Quadratic Unconstrained Binary Optimization...

https://arxiv.org/abs/2607.02622

9 **CONF**

Fine-Grained Computation Offload for Off-the-Shelf Servers in Tens of Lines

ArXiv CS.AI 50%

arXiv:2607.02630v1 Announce Type: cross Abstract: Hardware accelerators now sit on the critical path of online serving. GPUs, FPGAs, and increasingly remote services such as hardware security modules, post-quantum KEMs, and inference servers. For fine-grained offloads...

https://arxiv.org/abs/2607.02630

0 CONF

Walrus: A Cross-Domain Foundation Model for Continuum Dynamics

ArXiv CS.AI 50%

arXiv:2511.15684v2 Announce Type: replace-cross Abstract: Foundation models have transformed machine learning for language and vision, but achieving comparable impact in physical simulation remains a challenge. Data heterogeneity and unstable long-term dynamics inhibit learning...

https://arxiv.org/abs/2511.15684

1 CONF

Three-Phase Evaluation of AI-Assisted Software Development Life Cycle

ArXiv CS.AI 50%

arXiv:2607.05125v1 Announce Type: cross Abstract: This paper presents an exploratory evaluation of how increasing levels of AI autonomy affect software development productivity, requirement adherence, and developer cognitive workload. A team of four developers reimplemented the...

https://arxiv.org/abs/2607.05125

2 CONF

Not All Refusals Are Equal: How Safety Alignment Fails Cybersecurity at Scale

ArXiv CS.AI 50%

arXiv:2607.02714v1 Announce Type: cross Abstract: There is no doubt that safety alignment is an essential step in LLM training. However, conceptually it does not distinguish between various domains and the level of potential harm of a query, which creates significant...

https://arxiv.org/abs/2607.02714

3 CONF

A harmonised dataset for Earth system foundation models

ArXiv CS.AI 50%

arXiv:2607.03298v1 Announce Type: cross Abstract: Foundation models for Earth systems have so far been trained primarily on physical climate and weather data, with limited representation of the human systems that both drive and respond to environmental change. The lack of a...

https://arxiv.org/abs/2607.03298

4 CONF

One Prompt, Many Sounds: Modeling Listener Variability in LLM-Based Equalization

ArXiv CS.AI 50%

arXiv:2601.09448v3 Announce Type: replace-cross Abstract: Conventional audio equalization is a static process that requires manual and cumbersome adjustments to adapt to changing listening contexts (e.g., mood, location, or social setting). In this paper, we introduce a Large...

https://arxiv.org/abs/2601.09448

5 CONF

Self-Improving Diffusion Classifiers with Minority Preference Optimization

ArXiv CS.AI 50%

arXiv:2607.03770v1 Announce Type: cross Abstract: Prior studies have demonstrated that diffusion classifiers achieve robust zero-shot classification performance. However, their effectiveness is strongly tied to the pretraining data distribution: they perform well in majority,...

https://arxiv.org/abs/2607.03770

6 **CONF**

StarTSE: Towards Streaming Target Speaker Extraction via Chunk-wise Interleaved Splicing of Autoregressive Language Model

ArXiv CS.AI **50%**

arXiv:2604.19635v2 Announce Type: replace-cross Abstract: While generative models have set new benchmarks for Target Speaker Extraction (TSE), their inherent reliance on global context precludes deployment in real-time applications. Direct adaptation to streaming scenarios often...

https://arxiv.org/abs/2604.19635

7 **CONF**

JavaVulBench: A Java Vulnerability Benchmark with Realistic Splits, a Unified Multi-Backend Harness, and a Leakage-Aware Evaluation Mode

ArXiv CS.AI **50%**

arXiv:2607.02825v1 Announce Type: cross Abstract: We release \textsc{JavaVulBench}, a benchmark dataset and evaluation harness for Java vulnerability detection. The dataset contains $\sim$30{,}600 Java methods spanning 1{,}740 CVEs and 700+ projects, labelled at both method and...

https://arxiv.org/abs/2607.02825

8 **CONF**

Determinants and Limits of LLM Security-Tool Orchestration: A Study with HexStrike-AI

ArXiv CS.AI **50%**

arXiv:2607.02873v1 Announce Type: cross Abstract: Large language model agents driving security tool suites over the Model Context Protocol are increasingly common. Yet the factors that bound their capability remain poorly characterized: how much depends on the model versus the...

https://arxiv.org/abs/2607.02873

9 **CONF**

Where do LLMs Fall Short in CBT-Guided Affective Reasoning?

ArXiv CS.AI **50%**

arXiv:2607.02885v1 Announce Type: cross Abstract: Cognitive Behavioral Therapy (CBT) provides a structured framework for understanding a user's mental state by examining the interaction between cognitive and behavioral factors. However, out-of-the-box LLMs respond fluently and...

https://arxiv.org/abs/2607.02885

0 **CONF**

G2VD: Generalizable AI-Generated Video Detection via Counterfactual Intervention and Causal Disentanglement

ArXiv CS.AI **50%**

arXiv:2607.04607v1 Announce Type: cross Abstract: The rapid advancement of AI-generated videos poses increasing security risks and calls for robust detectors with strong cross-domain generalization. Although existing methods achieve promising results under in-domain evaluation,...

https://arxiv.org/abs/2607.04607

1 **CONF**

Learning Taxonomic Trees with Hierarchical Representation Regularization for Large Multimodal Models

ArXiv CS.AI **50%**

arXiv:2607.02909v1 Announce Type: cross Abstract: Taxonomies provide key information about the semantic relationships between concepts and the inherent organization of vision and language. Despite their impressive capabilities, large multimodal models (LMMs) often lack taxonomic...

https://arxiv.org/abs/2607.02909

2 CONF

Harmonic-Aware Transformer for Real-Time Catheter Localization in Interventional Procedures of Magnetic Particle Imaging

ArXiv CS.AI 50%

arXiv:2607.02919v1 Announce Type: cross Abstract: Magnetic particle imaging (MPI) enables real-time, radiation-free tracking of magnetic nanoparticle-coated instruments, making it highly suitable for interventional procedures. This study proposes a harmonic-aware transformer...

https://arxiv.org/abs/2607.02919

3 CONF

HCSU: A Dataset and Benchmark for Fine-Grained Historical Calligraphy Style Understanding

ArXiv CS.AI 50%

arXiv:2607.04147v1 Announce Type: cross Abstract: Automated fine-grained perception of calligraphy styles--a task vital to cultural heritage preservation--remains a critical challenge for Large Vision-Language Models (LVLMs), largely constrained by existing datasets that suffer...

https://arxiv.org/abs/2607.04147

4 CONF

The Foreign Policy AI Evaluation Gap

ArXiv CS.AI 50%

arXiv:2607.02955v1 Announce Type: cross Abstract: We argue that AI systems used in conducting foreign policy tasks - broadly enacting 'statecraft' - should be a priority test case for technical AI governance research. In enacting foreign policy, we refer to the formulation and...

https://arxiv.org/abs/2607.02955

5 CONF

Individual Parameters in Weight-Sparse Transformers Appear Interpretable

ArXiv CS.AI 50%

arXiv:2607.02964v1 Announce Type: cross Abstract: A central goal of mechanistic interpretability is to understand how neural networks work and what each individual component does. Dominant circuit-finding approaches focus on a specific behavior and reverse-engineer the role of...

https://arxiv.org/abs/2607.02964

6 CONF

CONFLUX: A Latent Diusion Model for 3D Chest-CT Synthesis with RL Post-Training

ArXiv CS.AI 50%

arXiv:2607.02998v1 Announce Type: cross Abstract: Controllable generative models of 3D medical images can synthesize volumes with specified clinical attributes, but this demands samples that are simultaneously high-fidelity, natively 3D, and faithful to the requested...

https://arxiv.org/abs/2607.02998

7 CONF

PosterHarness: Turning Scientific Poster Generation into an Auditable Instruction-Following Benchmark

ArXiv CS.AI 50%

arXiv:2607.03006v1 Announce Type: cross Abstract: Text-rich image models can now design poster-scale layouts, but we lack ways to measure whether they honor scientific communication contracts: legible labels, prescribed aspect ratios, and -- above all -- abstaining from...

https://arxiv.org/abs/2607.03006

8 **CONF**

EmCom-Diffusion: Probing Visual Reflection in Emergent Languages via Image Generation

ArXiv CS.AI **50%**

arXiv:2607.03752v1 Announce Type: cross Abstract: Measuring the extent to which emergent languages encode the visual content of their inputs is an open problem. We refer to this property as visual reflection: the extent to which emergent messages preserve information about their...

https://arxiv.org/abs/2607.03752

9 **CONF**

HyperVAttention: Efficient Sparse Attention with Spatio-Temporal Clustering for Video Diffusion

ArXiv CS.AI **50%**

arXiv:2607.03012v1 Announce Type: cross Abstract: Video Diffusion Transformers (VDiTs) have demonstrated significant capabilities in high-fidelity video generation. However, their ability to produce long-duration videos is fundamentally constrained by the quadratic complexity of...

https://arxiv.org/abs/2607.03012

0 **CONF**

LACE-SVD: Loss-Aware SVD with Cumulative Error Correction for LLM Compression

ArXiv CS.AI **50%**

arXiv:2607.03057v1 Announce Type: cross Abstract: The rapid growth in the parameter scale of large language models (LLMs) has created a strong demand for efficient compression techniques. As a hardware-agnostic and highly compatible approach, low-rank compression has been widely...

https://arxiv.org/abs/2607.03057

1 **CONF**

STELLA: Efficient Sensor-to-LLM Translation for On-Device Human Activity Recognition

ArXiv CS.AI **50%**

arXiv:2607.03089v1 Announce Type: cross Abstract: HAR is increasingly expected to run continuously on edge devices, yet recent LLM-based methods remain hard to deploy: raw sensor prompts are long, cloud inference adds latency and privacy risk, and fine-tuned LLM pipelines turn...

https://arxiv.org/abs/2607.03089

2 **CONF**

Don't Wait to Reply: Towards Responsive yet Thoughtful Dialogue through Proactive Thinking

ArXiv CS.AI **50%**

arXiv:2607.03093v1 Announce Type: cross Abstract: Thinking has emerged as a critical capability for Large Language Models (LLMs) tackling complex tasks. However, its reactive nature, where reasoning is passively triggered only upon receiving a user response, inevitably...

https://arxiv.org/abs/2607.03093

3 **CONF**

Adaptive Inference Batching using Policy Gradients

ArXiv CS.AI **50%**

arXiv:2607.05272v1 Announce Type: cross Abstract: Inference serving systems must balance throughput and latency under bursty, heterogeneous workloads, yet the industry standard remains static batching policies that require manual tuning and cannot adapt to shifting traffic. We...

https://arxiv.org/abs/2607.05272

4 **CONF**

ACPO: Adaptive Credit Policy Optimization via Fine-Grained Surrogate Entropy

ArXiv CS.AI **50%**

arXiv:2607.03126v1 Announce Type: cross Abstract: Reinforcement Learning (RL) has substantially improved the reasoning ability of large language models (LLMs), but sparse outcome rewards still make token-level credit assignment difficult. Existing scalable RL methods typically...

https://arxiv.org/abs/2607.03126

5 **CONF**

A Multi-Task Deep Learning Framework for Real-Time Intelligent Video Surveillance with Temporal Event Validation

ArXiv CS.AI **50%**

arXiv:2607.03131v1 Announce Type: cross Abstract: Modern video surveillance systems generate far more video streams than human operators can effectively monitor, making automated analysis essential for timely detection of security events. This paper presents a unified multi-task...

https://arxiv.org/abs/2607.03131

6 **CONF**

Text as Partial Constraint: Core-Residual Alignment for Robust Vision-Language Learning

ArXiv CS.AI **50%**

arXiv:2607.03143v1 Announce Type: cross Abstract: Vision-language alignment powers open-vocabulary recognition, retrieval, and LVLM grounding, yet natural captions are often underspecified, making similarity brittle and overly confident under paraphrase and omitted details. We...

https://arxiv.org/abs/2607.03143

7 **CONF**

Agentic-V2X: Small Language Model Agents for Deadline-Aware V2X Scheduling in 5G/6G Networks

ArXiv CS.AI **50%**

arXiv:2607.04290v1 Announce Type: cross Abstract: Large Language Models (LLMs) are proposed as control interfaces for next-generation networks, but their latency, hallucinations, and lack of control guarantees make them unsuitable for near-real-time packet schedulers, especially...

https://arxiv.org/abs/2607.04290

8 **CONF**

The Role of Prompt Language and Translation-Theory-Driven Prompts in Large Language Models: A Case Study on Spanish-Chinese Journalistic Translation

ArXiv CS.AI **50%**

arXiv:2607.03160v1 Announce Type: cross Abstract: This study examines how prompt language and translation theory-driven prompt design influence the quality of Spanish-Chinese journalistic translations generated by GPT-5.2. A parallel corpus of four editorials from El Pais was...

https://arxiv.org/abs/2607.03160

9 **CONF**

KARMA: Knowledge graph-based Automated Reasoning Materialization and Alignment

ArXiv CS.AI **50%**

arXiv:2607.03166v1 Announce Type: cross Abstract: Template-based contrastive synthesis is scalable, but its candidates often differ only in a few entity-slots while sequence-level optimization spreads supervision over mostly shared templates. We formalize this as the Resolution...

https://arxiv.org/abs/2607.03166

00 **CONF**

Effectiveness of LLM-based Software Diversity for Reliability Improvement -- an Empirical Study

ArXiv CS.AI **50%**

arXiv:2607.03174v1 Announce Type: cross Abstract: Software diversity has been extensively studied as a means of reducing the risk of common-mode failures. Classic work showed that the central issue is whether failures of diversely redundant components overlap in ways that limit...

https://arxiv.org/abs/2607.03174

01 **CONF**

Teaming Up with AI: Coordination and Cooperation

ArXiv CS.AI **50%**

arXiv:2607.03181v1 Announce Type: cross Abstract: Successful diffusion of AI in the workforce hinges on the economic value that AI brings to human endeavors. Bringing AI into the workforce is more than deploying a powerful new technology -- it is launching a new form of...

https://arxiv.org/abs/2607.03181

02 **CONF**

Transition Information Density: Morphological Trajectories, Synesthetic Perception, and Structured Interpolation in Neural Training (or: The Synesthetic AI)

ArXiv CS.AI **50%**

arXiv:2607.03210v1 Announce Type: cross Abstract: Standard machine learning training presents data as discrete endpoint pairs, omitting the structure of the space between them. This paper introduces Transition Information Density (TID) -- the information content recoverable from...

https://arxiv.org/abs/2607.03210

03 **CONF**

OpenGlass: A Sensing-Computing Split Architecture for Local MLLM-Driven Real-Time Visual Assistance

ArXiv CS.AI **50%**

arXiv:2607.03213v1 Announce Type: cross Abstract: We present OpenGlass, an open-source, privacy-oriented, local-first system for low-latency multimodal visual assistance, with a primary focus on blind and low-vision users. Cloud MLLM assistants offer strong visual understanding,...

https://arxiv.org/abs/2607.03213

04 **CONF**

Builder, Defender, Breaker: The Case Against Removing the Human from the AI-Driven Security Lifecycle

ArXiv CS.AI **50%**

arXiv:2607.03215v1 Announce Type: cross Abstract: Artificial intelligence has spread across the whole of the security lifecycle. The same family of models now writes application code, hardens it, and probes it for weaknesses, so that a single generative substrate increasingly...

https://arxiv.org/abs/2607.03215

05 **CONF**

Agentic and Generative AI for Open-Source Intelligence and Cyber Investigations: Taxonomy, Evaluation, Challenges, and Future Directions

ArXiv CS.AI **50%**

arXiv:2607.03233v1 Announce Type: cross Abstract: The rapid growth of publicly available digital information has rendered manual open-source intelligence (OSINT) analysis insufficient for modern intelligence, cybersecurity, and cyber investigation. Large language models (LLMs)...

https://arxiv.org/abs/2607.03233

06 **CONF**

PedestrianDiffusion: Multimodal Generative Denoising and Dense State Estimation for Inertial Navigation

ArXiv CS.AI **50%**

arXiv:2607.03349v1 Announce Type: cross Abstract: The accuracy of consumer-grade inertial navigation is bottlenecked by the stochastic noise of Micro-Electro-Mechanical Systems (MEMS). Traditional deterministic neural architectures often succumb to ``estimation jittering,''...

https://arxiv.org/abs/2607.03349

07 CONF

Brand-as-Memory: Vision-Language Models Encode Causal, Mechanistically Localizable Credibility Priors for News Sources

ArXiv CS.AI 50%

arXiv:2607.03365v1 Announce Type: cross Abstract: Vision-language models (VLMs) increasingly read news and web content as images, where the publisher's identity is visually present. We show that VLMs carry a strong source-credibility prior keyed on outlet identity, and study it...

https://arxiv.org/abs/2607.03365

08 CONF

Amortising Bayesian Experimental Design for Sequential Information Gathering in LLMs

ArXiv CS.AI 50%

arXiv:2607.03426v1 Announce Type: cross Abstract: Large language models (LLMs) exhibit strong reasoning and world-knowledge capabilities, yet often struggle to gather information effectively across the multi-turn interactions required in sequential decision-making settings. We...

https://arxiv.org/abs/2607.03426

09 CONF

No Time Like the Present: Agentic Test-Time Training for LLM Agents

ArXiv CS.AI 50%

arXiv:2607.03441v1 Announce Type: cross Abstract: LLM agents often degrade over long episodes: as trajectories grow, they revisit explored states, repeat failed actions, and lose strategies that previously worked. Test-time training (TTT) offers a way to adapt model weights to...

https://arxiv.org/abs/2607.03441

10 CONF

CaresAI at SMM4H-HeaRD 2026: Predicting TNM Staging

ArXiv CS.AI 50%

arXiv:2607.03466v1 Announce Type: cross Abstract: This study aims to predict Tumor, Node, and Metastasis (TNM) stage labels independently, with the Cancer Genome Atlas (TCGA) pathology report as the sixth shared task of SMM4H-HeaRD 2026. The problem is framed as three...

https://arxiv.org/abs/2607.03466

11 CONF

Towards Diverse and Comprehensive Benchmarks for Mutual Information Estimation

ArXiv CS.AI 50%

arXiv:2607.03487v1 Announce Type: cross Abstract: Mutual information (MI) estimation is a central problem in machine learning and statistics; however, existing benchmarks typically evaluate estimators on simplified, low-dimensional distributions, leaving their performance on...

https://arxiv.org/abs/2607.03487

12 CONF

CAGE-1: Control, Assurance, and Governance Evaluation for Enterprise Agentic AI

ArXiv CS.AI 50%

arXiv:2607.03510v1 Announce Type: cross Abstract: Enterprise artificial intelligence is moving from experimentation into operational workflows. Early programs focused on model access and retrieval-augmented generation, but enterprises are now beginning to deploy agents that...

https://arxiv.org/abs/2607.03510

13 CONF

AGL-1: The Enterprise AI Governance Layer as a Control Plane for Trusted Enterprise Intelligence

ArXiv CS.AI 50%

arXiv:2607.03516v1 Announce Type: cross Abstract: Enterprise artificial intelligence is moving from isolated experimentation toward operational dependency across copilots, retrieval-augmented generation systems, autonomous agents, and AI-enabled business workflows. As this...

https://arxiv.org/abs/2607.03516

14 CONF

PLGSA-Transformer: Periocular Landmark-Guided Attention with Occlusion-Adaptive Cosine Thresholding for Cross-Modal Masked and Unmasked Face Recognition

ArXiv CS.AI 50%

arXiv:2607.03581v1 Announce Type: cross Abstract: The widespread adoption of facial masks, accelerated by COVID-19 and mandated in security-sensitive settings, has exposed limitations of conventional face recognition systems. Existing approaches relying on fixed cosine...

https://arxiv.org/abs/2607.03581

15 CONF

An AI-Assisted Solution to the Signed BAR Conjecture: Uniqueness in the Harrison--Reiman Class and a Completely-$\mathcal{S}$ Class Obstruction

ArXiv CS.AI 50%

arXiv:2607.03639v1 Announce Type: cross Abstract: For a multidimensional reflected diffusion, determining whether the associated basic adjoint relationship (BAR) uniquely characterizes the stationary distribution is a basic uniqueness problem in the BAR approach. The problem has...

https://arxiv.org/abs/2607.03639

16 CONF

ViPo-MLLM: Visual-Pose Multimodal LLM for Gloss-Free Sign Language Translation

ArXiv CS.AI 50%

arXiv:2607.03657v1 Announce Type: cross Abstract: Gloss-free Sign Language Translation (SLT) translates sign language videos into spoken-language sentences without gloss annotations, avoiding costly labeling but requiring fine-grained modeling of hands, body, and facial cues....

https://arxiv.org/abs/2607.03657

17 CONF

Phase-Preserving Trimodal Transformer for Tropical Forest Biomass Estimation Using Optical and PolInSAR Data

ArXiv CS.AI 50%

arXiv:2607.03663v1 Announce Type: cross Abstract: The accurate estimation of Above-Ground Biomass (AGB) in mature tropical forests remains a critical challenge in remote sensing, primarily due to the saturation of Synthetic Aperture Radar (SAR) signals in high-density areas and...

https://arxiv.org/abs/2607.03663

18 CONF

CoGen3D: An Agentic Human-AI Co-Design Pipeline for 3D Asset Generation for Virtual Reality

ArXiv CS.AI 50%

arXiv:2607.03731v1 Announce Type: cross Abstract: Creating 3D assets for virtual reality requires modeling expertise, which restricts the authorship of immersive experiences. Existing generative AI tools rely on unconstrained, command-driven prompting, lacking the conversational...

https://arxiv.org/abs/2607.03731

19 **CONF**

Rethinking Depth Pruning for Vision Transformers: A Heterogeneity-Aware Perspective

ArXiv CS.AI **50%**

arXiv:2607.03784v1 Announce Type: cross Abstract: While prior studies have successfully compressed vision Transformers (ViTs) through various pruning techniques, most have concentrated on width pruning to achieve significant reductions in model size. Depth pruning, which removes...

https://arxiv.org/abs/2607.03784

20 **CONF**

Hyperparameter Transfer in Graph Neural Networks

ArXiv CS.AI **50%**

arXiv:2607.05017v1 Announce Type: cross Abstract: The performance of deep learning models crucially depends on the settings of hyperparameters like learning rate, initialization scale, and weight decay. Hyperparameter transfer aims to make near-optimal hyperparameter settings...

https://arxiv.org/abs/2607.05017

21 **CONF**

Punching Above Their Weight: Classification-Head Fine-Tuning of Tiny Language Models (TLMs) for Verifiable Multiple-Choice Tasks

ArXiv CS.AI **50%**

arXiv:2607.03801v1 Announce Type: cross Abstract: We define Tiny Language Models (TLMs) as models below roughly 3B parameters that fit on mainstream consumer devices. We study how to adapt them for and use them on verifiable multiple-choice tasks. We compare three LoRA-based...

https://arxiv.org/abs/2607.03801

22 **CONF**

CineMobile: On-Device Image-to-Video Diffusion for Cinematic Camera Motion Generation

ArXiv CS.AI **50%**

arXiv:2607.03803v1 Announce Type: cross Abstract: The growing demand for image-to-video creation on mobile devices has increasingly focused on cinematic motion effects like bullet time, dolly zoom, slow motion, etc. While Diffusion Transformers (DiTs) exhibit strong performance...

https://arxiv.org/abs/2607.03803

23 **CONF**

Scalable Semantic Steering of Embedding Projections

ArXiv CS.AI **50%**

arXiv:2607.03978v1 Announce Type: cross Abstract: Low-dimensional projections support interactive visual analysis of high-dimensional data embeddings, but their structure often does not align with analyst-defined semantic relationships. Recent LLM-augmented semantic steering...

https://arxiv.org/abs/2607.03978

24 **CONF**

How Do Diffusion Classifiers Decide? A Bias-Centric Evaluation

ArXiv CS.AI **50%**

arXiv:2607.03831v1 Announce Type: cross Abstract: Diffusion models have recently been repurposed for zero-shot classification, giving rise to diffusion classifiers that identify the best-matching text prompt by minimizing the noise-prediction error. Despite their growing...

https://arxiv.org/abs/2607.03831

25 **CONF**

High-Fidelity One-Step Generative Visuomotor Policy via Recursive Correction, Frequency Consistency, and Contrastive Flow Matching

ArXiv CS.AI **50%**

arXiv:2607.03865v1 Announce Type: cross Abstract: Generative models such as diffusion and flow matching have advanced robotic visuomotor policies by modeling multimodal action distributions, but their multi-step sampling or ODE solving introduces inference latency. Existing...

https://arxiv.org/abs/2607.03865

26 **CONF**

Next-Gen Sponsored Search: Crafting the Perfect Query with Inventory-Aware RAG (InvAwr-RAG) Based GenAI

ArXiv CS.AI **50%**

arXiv:2607.03880v1 Announce Type: cross Abstract: Sponsored search plays a crucial role in e-commerce revenue generation, where advertisers strategically bid on keywords to capture the attention of users through relevant search queries. However, the process of identifying...

https://arxiv.org/abs/2607.03880

27 **CONF**

Enhancement of E-commerce Sponsored Search Relevancy with LLM

ArXiv CS.AI **50%**

arXiv:2607.03886v1 Announce Type: cross Abstract: Sponsored search plays a crucial role as a revenue stream for search engines, wherein advertisers competitively bid on keywords that align with the users' search queries. The task of matching relevant keywords to these queries is...

https://arxiv.org/abs/2607.03886

28 **CONF**

Probe, Don't Prompt: A Hidden-State Probe for Metadata Filtering in Multi-Meta-RAG

ArXiv CS.AI **50%**

arXiv:2607.03929v1 Announce Type: cross Abstract: Multi-Meta-RAG improves retrieval for multi-hop question answering by filtering a vector store on metadata (the news source) that it extracts from each query by prompting gpt-3.5-turbo. We show this proprietary, free-form...

https://arxiv.org/abs/2607.03929

29 **CONF**

MPSelectTune: Prompt-type Selection for Fine-tuning improves Concept Unlearning in LLMs

ArXiv CS.AI **50%**

arXiv:2607.03932v1 Announce Type: cross Abstract: LLMs can be conveniently adapted to a diverse set of tasks, e.g, prediction, question-answering tasks, etc, using appropriate prompts with few-shot examples. Biased or harmful concepts, e.g. gender or bio-weapons, present in...

https://arxiv.org/abs/2607.03932

30 **CONF**

The Remarkable Effectiveness of Providing AI Agents with Natural Language Tools: A Replication Study Validating NLT Performance Across 14 Models

ArXiv CS.AI **50%**

arXiv:2607.03953v1 Announce Type: cross Abstract: This study independently replicates and extends the Natural Language Tools (NLT) framework of Johnson et al.~(2025), which questions the use of structured tool calling in large language model (LLM) agentic systems. We evaluated...

https://arxiv.org/abs/2607.03953

31 **CONF**

Refused in Chat, Written in Code: Workflow-Level Jailbreak Construction in IDE Coding Agents

ArXiv CS.AI 50%

arXiv:2607.03968v1 Announce Type: cross Abstract: Large language models are increasingly deployed as IDE-integrated coding agents that decompose tasks, generate and edit files, run code, and refine outputs over many turns. Yet their safety is still often evaluated as if they...

https://arxiv.org/abs/2607.03968

32 **CONF**

When Does Small Data Work? Accuracy and Efficiency Trade-offs Between Tabular Foundation Models and Conventional Methods for Crowd-State Classification at Hajj and Umrah

ArXiv CS.AI 50%

arXiv:2607.04013v1 Announce Type: cross Abstract: Learning from few labeled examples is a central challenge in tabular machine learning, and it becomes the binding constraint in domains where labeling is costly, such as crowd monitoring during Hajj and Umrah. Tabular foundation...

https://arxiv.org/abs/2607.04013

33 **CONF**

OmniOpt: Taxonomy, Geometry, and Benchmarking of Modern Optimizers

ArXiv CS.AI 50%

arXiv:2607.04033v1 Announce Type: cross Abstract: Optimizer selection for large-scale model training has become a system-level design decision constrained jointly by compute, memory, tuning budget, and task diversity, yet the landscape of over one hundred methods remains...

https://arxiv.org/abs/2607.04033

34 **CONF**

Telescope: Improving Zero Shot Detection of LLM Generated Content By Measuring Token Repetition Probability

ArXiv CS.AI 50%

arXiv:2607.04061v1 Announce Type: cross Abstract: Distinguishing Large Language Model (LLM) generated text from human writing is a critical and difficult challenge. While LLMs are trained to write like humans, we hypothesize that this training leaves an indelible mark. LLMs...

https://arxiv.org/abs/2607.04061

35 **CONF**

Beyond Multilingual Averages: MTEB-PT, a Benchmark for Portuguese Sentence Encoders

ArXiv CS.AI 50%

arXiv:2607.04071v1 Announce Type: cross Abstract: Portuguese remains underrepresented in text embedding evaluation, despite being one of the most widely spoken languages in the world. As a result, embedding models are often selected based on English or multilingual metrics,...

https://arxiv.org/abs/2607.04071

36 **CONF**

Benchmarking API Drift in LLM-Generated Quantum Code Across Successive SDK Versions

ArXiv CS.AI 50%

arXiv:2607.04072v1 Announce Type: cross Abstract: Large language models can generate plausible quantum code, but it is unclear whether they can reliably target the specific software development kit (SDK) version requested by the user. We study this problem as API drift and...

https://arxiv.org/abs/2607.04072

37 **CONF**

CSB: A Counting and Sampling tool for Bit-vectors

ArXiv CS.AI **50%**

arXiv:2607.04142v1 Announce Type: cross Abstract: Satisfiability modulo theory (SMT) solvers have significantly advanced automated reasoning due to their effectiveness in solving problems across various fields. With the advancement in SMT solvers, there is growing interest in...

https://arxiv.org/abs/2607.04142

38 **CONF**

Information-Geometric Superposed Vowel Evaluation: Part 1. Moraic Syllabary (Japanese)

ArXiv CS.AI **50%**

arXiv:2607.04154v1 Announce Type: cross Abstract: This paper explains the principles and provides examples of a new method for distinguishing between FAKE human speech synthesized by generative AI and natural speech. Since synthetic speech is generated based on information from...

https://arxiv.org/abs/2607.04154

39 **CONF**

SeeMe: Mitigating Hallucinations in Large Vision-Language Models through Effective Visual Token Engineering

ArXiv CS.AI **50%**

arXiv:2607.04163v1 Announce Type: cross Abstract: Large Vision-Language Models (LVLMs) have achieved remarkable progress in visual understanding tasks such as image captioning and visual question answering. However, they remain susceptible to hallucinations, generating content...

https://arxiv.org/abs/2607.04163

40 **CONF**

CritiqueDriveVLM: From Verifier-Guided Reinforcement Learning to Latent Thought Distillation for Autonomous Driving

ArXiv CS.AI **50%**

arXiv:2607.04179v1 Announce Type: cross Abstract: End-to-end Vision-Language Models (VLMs) show immense potential in autonomous driving. However, standard Supervised Fine-Tuning (SFT) often suffers from reasoning hallucinations and conservative biases. While traditional...

https://arxiv.org/abs/2607.04179

41 **CONF**

Signal or Noise? Understanding Generative Models for Real-World Sensor Time Series

ArXiv CS.AI **50%**

arXiv:2607.04245v1 Announce Type: cross Abstract: Generative models have changed how machine learning represents complex data distributions, especially in language and vision, yet many real-world systems are observed instead as continuous, high-dimensional, and noisy sensor time...

https://arxiv.org/abs/2607.04245

42 **CONF**

Large Language Models Develop Novel Social Biases Through Adaptive Exploration

ArXiv CS.AI **50%**

arXiv:2511.06148v4 Announce Type: replace-cross Abstract: As large language models (LLMs) are adopted into frameworks that grant them the capacity to make real decisions, it is increasingly important to ensure that they are unbiased. In this paper, we argue that the predominant...

https://arxiv.org/abs/2511.06148

43 CONF

Risk-Constrained Freshness-Aware Semantic Caching for Open-Web Retrieval-Augmented LLMs

ArXiv CS.AI 50%

arXiv:2607.04281v1 Announce Type: cross Abstract: Semantic caching reduces the latency and cost of retrieval-augmented generation (RAG) by serving cached answers to semantically similar queries, but most existing methods do not model the time-varying freshness of open-web...

https://arxiv.org/abs/2607.04281

44 CONF

CausalGame: Benchmarking Causal Thinking of LLM Agents in Games

ArXiv CS.AI 50%

arXiv:2607.04293v1 Announce Type: cross Abstract: Building AI Scientist agents with Large Language Models (LLMs) has recently attracted growing attention. Since scientific discovery fundamentally relies on uncovering causal relationships from observations, the capability of...

https://arxiv.org/abs/2607.04293

45 CONF

HiFA4: Training-Free 4-bit FlashAttention on Ascend HIF4 NPUs for LLM Inference

ArXiv CS.AI 50%

arXiv:2607.04302v1 Announce Type: cross Abstract: We present HiFA4, a post-training operator-level design that executes both QK^T and PV in FlashAttention as 4-bit HIF4 Cube GEMMs for LLM inference on Ascend NPUs, while maintaining the online softmax state in FP16. To our...

https://arxiv.org/abs/2607.04302

46 CONF

Full-Stack FP4: Stable LLM Pretraining with Quantized Projections, Optimizers, and Attention

ArXiv CS.AI 50%

arXiv:2607.04422v1 Announce Type: cross Abstract: Recent NVFP4 pretraining methods mainly target transformer linear layers, leaving optimizer states, optimizer arithmetic and attention underexplored in 4-bit pipelines. This critical gap blocks stable full-stack 4-bit...

https://arxiv.org/abs/2607.04422

47 CONF

Transferability Between Understanding and Generation in Unified Multimodal Models

ArXiv CS.AI 50%

arXiv:2607.04423v1 Announce Type: cross Abstract: Unified Multimodal Models (UMMs) integrate image understanding and generation within a single architecture, yet how the two tasks interact remains understudied. We investigate $\boldsymbol{\mathsf{transferability}}$ in UMMs:...

https://arxiv.org/abs/2607.04423

48 CONF

BoRP: Bootstrapped Regression Probing for Scalable and Human-Aligned LLM Evaluation

ArXiv CS.AI 50%

arXiv:2601.18253v2 Announce Type: replace-cross Abstract: Accurate evaluation of user satisfaction is critical for iterative development of conversational AI. However, for open-ended assistants, traditional A/B testing lacks reliable metrics: explicit feedback is sparse, while...

https://arxiv.org/abs/2601.18253

49 **CONF**

Covert Trait Propagation Is Representation Alignment: Mechanistic Evidence from Hidden-Channel Distillation

ArXiv CS.AI **50%**

arXiv:2607.04432v1 Announce Type: cross Abstract: A student model trained on pure uniform noise can still inherit its teacher's digit-classification ability, provided the two share initialization. Previous work proves this transfer is guaranteed when the teacher's learning rate...

https://arxiv.org/abs/2607.04432

50 **CONF**

DSWAM: A Dual-System World Action Foundation Model for Fine-Grained Robot Manipulation

ArXiv CS.AI **50%**

arXiv:2607.04927v1 Announce Type: cross Abstract: World Action Models (WAMs) provide a promising alternative to Vision-Language-Action (VLA) policies by using video-based world modeling as dense supervision for robot action learning. Existing WAMs excel at physically grounded...

https://arxiv.org/abs/2607.04927

51 **CONF**

Generative wave propagator

ArXiv CS.AI **50%**

arXiv:2607.04440v1 Announce Type: cross Abstract: Seismic wavefield simulation is fundamental to seismology, but conventional finite-difference (FD) methods remain limited by numerical dispersion and stability constraints, which often require dense spatial grids and small time...

https://arxiv.org/abs/2607.04440

52 **CONF**

A Deep Learning-based surrogate model for Severe Accidents in nuclear reactors using ASTEC

ArXiv CS.AI **50%**

arXiv:2607.04450v1 Announce Type: cross Abstract: Integral codes like the Accident Source Term Evaluation Code (ASTEC) are powerful tools to study the physics of Severe Accidents (SAs) in nuclear reactors. Real time SA simulators can also be helpful in training operators of...

https://arxiv.org/abs/2607.04450

53 **CONF**

PulmoSight-XAI: An Explainable Multi-View Attention Ensemble with Gradient Boosting Meta-Learning for Multi-Label Chest X-Ray Classification

ArXiv CS.AI **50%**

arXiv:2607.04478v1 Announce Type: cross Abstract: Automated chest X-ray classification remains challenging due to severe class imbalance, co-occurring pathologies, and the loss of localized features in conventional architectures. To address these, we propose an explainable...

https://arxiv.org/abs/2607.04478

54 **CONF**

Transplanting, inverting, and preventing a misalignment persona: method-conditional emergent misalignment in Qwen2.5

ArXiv CS.AI **50%**

arXiv:2607.04510v1 Announce Type: cross Abstract: Emergent misalignment (EM) -- the broad misbehaviour a language model acquires after fine-tuning on narrow harmful data -- is mediated in Qwen2.5 models by a latent persona direction, and that direction is causal in open weights....

https://arxiv.org/abs/2607.04510

55 CONF

Language Models Represent and Transform Concepts with Shared Geometry

ArXiv CS.AI 50%

arXiv:2607.04525v1 Announce Type: cross Abstract: How concepts are represented in neural networks is a fundamental question in machine learning. The dominant view treats concept representations as stationary geometric objects. Yet concepts appear in context, and context...

https://arxiv.org/abs/2607.04525

56 CONF

Training-Free Model Selection and Domain-Aware Score Calibration for First-Shot Anomalous Sound Detection

ArXiv CS.AI 50%

arXiv:2607.04526v1 Announce Type: cross Abstract: First-shot anomalous sound detection in DCASE Challenge Task 2 must flag anomalies of unseen machine types with a single threshold, without knowing whether a test clip comes from the data-rich source domain (990 normal training...

https://arxiv.org/abs/2607.04526

57 CONF

Lyapunov-Guided Training for Hardware-Safe Neural Networks Under Fixed-Point Arithmetic

ArXiv CS.AI 50%

arXiv:2607.04531v1 Announce Type: cross Abstract: Low-precision neural networks are attractive for resource-constrained hardware, but fixed-point arithmetic introduces failure modes that are often hidden by idealised quantisation models. In particular, two's-complement overflow...

https://arxiv.org/abs/2607.04531

58 CONF

Obey, Diverge, Collapse: Blind Obedience to Incorrect Instructions Drives Code LLMs to Irrecoverable Code Semantic Collapse

ArXiv CS.AI 50%

arXiv:2607.04537v1 Announce Type: cross Abstract: Code language models are now trusted collaborators in production workflows for debugging, refactoring, and iterative repair, and every benchmark that evaluates them assumes the instructions they act on are correct. We study what...

https://arxiv.org/abs/2607.04537

59 CONF

CRISP: A Spatiotemporal Camera-Radar Backbone for Driving via Forecasting-Based World-Model Pretraining

ArXiv CS.AI 50%

arXiv:2607.04541v1 Announce Type: cross Abstract: Camera-radar (CR) fusion is a practical sensing configuration for autonomous driving, but existing models are typically trained with task-specific supervision, limiting reusable representation learning. We present CRISP, a...

https://arxiv.org/abs/2607.04541

60 CONF

BluTrain: A C++/CUDA Framework for AI Systems

ArXiv CS.AI 50%

arXiv:2606.24780v2 Announce Type: replace Abstract: Progress in deep learning is, at scale, more a matter of systems engineering than of modelling: the behaviour of a model in training (its throughput, its memory footprint, and the numerical fidelity of the result) is determined...

https://arxiv.org/abs/2606.24780

61 **CONF**

Retroactive Chain-of-Thought (RetroCoT): Forensic Reconstruction Prompts as a Safety Diagnostic Across Model Generations

ArXiv CS.AI **50%**

arXiv:2607.04645v1 Announce Type: cross Abstract: Safety alignment in large language models is typically evaluated against direct, imperative harmful requests. We show that this alignment is highly conditioned on pragmatic register: models that refuse a direct request frequently...

https://arxiv.org/abs/2607.04645

62 **CONF**

Machine Learning for Depression Screening and Intervention: an Original Circadian Rhythm Score-based Methodology

ArXiv CS.AI **50%**

arXiv:2607.04648v1 Announce Type: cross Abstract: Depression screening from large-scale behavioral data is challenged by fragmented circadian indicators, limited interpretability, and the lack of intervention-oriented analysis. Existing approaches typically analyze sleep,...

https://arxiv.org/abs/2607.04648

63 **CONF**

Elastic Gang: Per-Token Membership Change for a Hard-Barriered LLM Inference Gang Co-Scheduled with OS Processes

ArXiv CS.AI **50%**

arXiv:2607.04668v1 Announce Type: cross Abstract: On-device LLM decoding is a hard-barriered CPU-SIMD computation that wants every core for milliseconds per token, while the rest of the OS wants those same cores continuously. A barriered gang cannot simply be dropped into a...

https://arxiv.org/abs/2607.04668

64 **CONF**

Do Vision-Language-Action Models Mean What They Say? On the Role of Faithfulness in Embodied Reasoning

ArXiv CS.AI **50%**

arXiv:2607.04681v1 Announce Type: cross Abstract: Embodied Chain-of-Thought has emerged as a promising mechanism to enhance robot decision-making and interpretability in black-box Vision-Language Action (VLA) models. However, whether this verbalized Chain-of-Thought truthfully...

https://arxiv.org/abs/2607.04681

65 **CONF**

Dashboard2Code: Evaluating Multimodal Models on Reconstructing Interactive Dashboards

ArXiv CS.AI **50%**

arXiv:2607.04727v1 Announce Type: cross Abstract: Automatic data visualization generation has advanced rapidly with multi-modal large language models, yet existing efforts largely focus on static charts and overlook the interactive dashboards commonly used for real-world data...

https://arxiv.org/abs/2607.04727

66 **CONF**

Turning Off-Policy Tokens On-Policy: A Plug-in Approach for Improving LLM Alignment

ArXiv CS.AI **50%**

arXiv:2607.04728v1 Announce Type: cross Abstract: Reinforcement learning (RL) post-training for large language models (LLMs) follows a efficient paradigm of "rollout then update", which inevitably results in off-policy training data. To resolve this, Importance sampling (IS) is...

https://arxiv.org/abs/2607.04728

67 CONF

Activation-Deactivation: A General Framework for Robust Post-hoc Explainable AI

ArXiv CS.AI 50%

arXiv:2510.01038v2 Announce Type: replace Abstract: Perturbation-based explainability methods face criticism due to their reliance on out-of-distribution mutants. This raises doubts about the quality of the explanations. In this paper, we introduce a novel forward pass paradigm,...

https://arxiv.org/abs/2510.01038

68 CONF

SeaAlert: Robust Severity Classification and LLM-Based Information Extraction for Noisy Maritime Distress Communications

ArXiv CS.AI 50%

arXiv:2604.14163v2 Announce Type: replace-cross Abstract: Maritime distress communications transmitted over very high frequency (VHF) radio are safety-critical voice messages used to report emergencies at sea. Under the Global Maritime Distress and Safety System (GMDSS), such...

https://arxiv.org/abs/2604.14163

69 CONF

HamQASBench: A Hamiltonian-Informed Diagnostic Benchmark for Evaluating Quantum Architecture Search

ArXiv CS.AI 50%

arXiv:2607.04845v1 Announce Type: cross Abstract: Quantum Architecture Search (QAS) automates the design of parameterized quantum circuits for variational quantum algorithms, yet existing benchmarks organize instances by molecular identity or qubit count -- criteria agnostic to...

https://arxiv.org/abs/2607.04845

70 CONF

Pretraining Curricula Enable Selective Fine-tuning

ArXiv CS.AI 50%

arXiv:2607.04846v1 Announce Type: cross Abstract: Transformers follow implicit curricula whereby some tasks are learned before others. However, how explicit pretraining curricula influence learning, generalization, and the selectivity of fine-tuning is unclear. This is important...

https://arxiv.org/abs/2607.04846

71 CONF

EventCoT: Event-centric Video Chain-of-thought for Reasoning Temporal Localization

ArXiv CS.AI 50%

arXiv:2607.04872v1 Announce Type: cross Abstract: Reasoning temporal localization (RTL) requires a model to generate an answer that itself contains the time interval supporting it, so high-level reasoning and precise temporal grounding must be produced jointly in a single...

https://arxiv.org/abs/2607.04872

72 CONF

Input Pathways Shape Few-Shot, Not Zero-Shot, Binding in Tiny Transformers: A Fully-Enumerable Study

ArXiv CS.AI 50%

arXiv:2607.04926v1 Announce Type: cross Abstract: How does the way information reaches a transformer -- as symbolic tokens, a clean per-factor "oracle" code, or an entangled perceptual vector -- shape whether it binds that information compositionally? We study ~6-10K-parameter...

https://arxiv.org/abs/2607.04926

73 CONF

Joint Velocity Slope Diffusion Prior for Structurally Constrained Velocity Model Building

ArXiv CS.AI 50%

arXiv:2607.04982v1 Announce Type: cross Abstract: High-resolution velocity models are crucial for reservoir characterization and subsurface delineation. However, the band limited nature of our surface recorded data limits resolution. Utilizing well measurements to enhance the...

https://arxiv.org/abs/2607.04982

74 CONF

Comparison of Loss Functions for Robust Deep Learning-based Echocardiography Segmentation when Learning with Partially Labelled Data from Multiple Domains

ArXiv CS.AI 50%

arXiv:2607.05008v1 Announce Type: cross Abstract: Echocardiography is the first imaging modality used for assessing cardiac function, and accurate segmentation of cardiac structures is essential for deriving biomarkers. However, the development of effective automated...

https://arxiv.org/abs/2607.05008

75 CONF

ImputeECG: Deep Learning Reconstruction of Complete 12-Lead Electrocardiograms from Incomplete Recordings for Cardiac Assessment

ArXiv CS.AI 50%

arXiv:2607.05009v1 Announce Type: cross Abstract: Complete digital 12-lead electrocardiograms (ECGs) are essential for AI-enabled cardiovascular assessment, yet many clinical ECG records, particularly those digitized from ECG images, remain incomplete because of short display...

https://arxiv.org/abs/2607.05009

76 CONF

Your Agent's Memories Are Not Its Own: Forged Reasoning Attacks on LLM Agent Memory and Defenses

ArXiv CS.AI 50%

arXiv:2607.05029v1 Announce Type: cross Abstract: Persistent memory has enabled large language model (LLM) agents to store factual knowledge, prior decisions, reasoning histories, tool usage information, and context. While this has improved the agent's functionality and...

https://arxiv.org/abs/2607.05029

77 CONF

LLM-Based Test Oracles: Source-of-Authority Taxonomy -- A Systematic Literature Review

ArXiv CS.AI 50%

arXiv:2607.05031v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used to produce test oracles, the part of a test that decides whether observed behavior is correct. Yet a clear account of where these oracles draw their authority is missing. Prior...

https://arxiv.org/abs/2607.05031

78 CONF

RUFNet: Query-Guided Support Mask Refinement and Uncertainty Fusion based on Hybrid Mamba for Few-Shot Brain Tumor Segmentation

ArXiv CS.AI 50%

arXiv:2607.05035v1 Announce Type: cross Abstract: Few-shot brain tumor segmentation remains challenging due to noisy support masks, inter-patient variations between support and query images, and the lack of pixel-wise confidence estimation. This study proposes RUFNet, a Hybrid...

https://arxiv.org/abs/2607.05035

79 CONF

Grokking Is Conditional and Fragile: A Fully-Tractable, Multi-Seed Study at 12K Parameters

ArXiv CS.AI 50%

arXiv:2607.05104v1 Announce Type: cross Abstract: Grokking -- the delayed onset of generalization long after a network has fit its training set - -is usually studied in models too large to read completely and reported from single training runs. We instead study a publicly...

https://arxiv.org/abs/2607.05104

80 CONF

Agent Data Injection Attacks are Realistic Threats to AI Agents

ArXiv CS.AI 50%

arXiv:2607.05120v1 Announce Type: cross Abstract: AI agents act on behalf of user prompts, consuming external data and taking actions based on the agent context. Prior research on AI agent security has primarily focused on indirect prompt injection (IPI). Its most well-studied...

https://arxiv.org/abs/2607.05120

81 CONF

PDEFlow: Autonomous Agentic PDE Pipelines for Neural Operator Learning and Solver-Free Inference

ArXiv CS.AI 50%

arXiv:2607.05134v1 Announce Type: cross Abstract: We present PDEFlow, an autonomous agentic framework that turns user-level ODE and PDE descriptions into solver-backed neural-operator pipelines. The workflow links problem specification, data generation, operator training, and...

https://arxiv.org/abs/2607.05134

82 CONF

Shifting from Discrete to Continuous Reference Data: QSM-Derived Horizontal Tree Biomass Distribution for Deep Learning Biomass Estimation

ArXiv CS.AI 50%

arXiv:2607.05260v1 Announce Type: cross Abstract: Conventional modeling approaches for LiDAR-based above-ground biomass (AGB) estimation rely on discrete plot-level inventory aggregates. This methodology introduces boundary-effect uncertainties that may severely degrade model...

https://arxiv.org/abs/2607.05260

83 CONF

ProPS: Prompted Profile Synthesis for Natural Language-Conditioned Speaker Embedding Distributions

ArXiv CS.AI 50%

arXiv:2607.05276v1 Announce Type: cross Abstract: Speaker embeddings, or x-vectors, are widely used to represent speaker identity and speaker-related attributes, but existing embedding extractors are typically descriptive rather than generative: they map an observed speech...

https://arxiv.org/abs/2607.05276

84 CONF

Air Quality Downscaling with Station-Guided Pseudo-Supervision

ArXiv CS.AI 50%

arXiv:2607.05292v1 Announce Type: cross Abstract: Super-resolving coarse atmospheric fields to local PM$_{2.5}$ variations is uniquely challenged by a mismatch in spatial support: while pixels represent regional averages, ground-truth observations are discrete, unaligned samples...

https://arxiv.org/abs/2607.05292

85 CONF

What Does a Discrete Diffusion Model Learn?

ArXiv CS.AI 50%

arXiv:2607.05381v1 Announce Type: cross Abstract: What does a discrete diffusion model learn: a denoiser, a score ratio, or a bridge plug-in predictor? At the level of jump rates, these are one object in different coordinates, and reading a neural network in the wrong coordinate...

https://arxiv.org/abs/2607.05381

86 CONF

AI's Blind Spots: Geographic Knowledge and Diversity Deficit in Generated Urban Scenario

ArXiv CS.AI 50%

arXiv:2506.16898v2 Announce Type: replace Abstract: Diffusion-based text-to-image models are increasingly used for urban analysis and scenario generation, but their geographic knowledge and representational biases remain poorly understood. We evaluate FLUX 1-schnell and Stable...

https://arxiv.org/abs/2506.16898

87 CONF

The Language of Bargaining: Linguistic Effects in LLM Negotiations

ArXiv CS.AI 50%

arXiv:2601.04387v2 Announce Type: replace Abstract: Negotiation is a core component of social intelligence, requiring agents to balance strategic reasoning, cooperation, and social norms. Recent work shows that LLMs can engage in multi-turn negotiation, yet nearly all...

https://arxiv.org/abs/2601.04387

88 CONF

Iterative Visual Thinking and the Self-Correction Mirage in VLM Grounding

ArXiv CS.AI 50%

arXiv:2606.13156v2 Announce Type: replace-cross Abstract: Letting a vision-language model (VLM) think longer at test time has driven much recent progress. A natural way to bring this to spatial grounding is visual self-correction: the model predicts a bounding box, sees it...

https://arxiv.org/abs/2606.13156

89 CONF

Framework of Thoughts: A Foundation Framework for Dynamic and Optimized Reasoning based on Chains, Trees, and Graphs

ArXiv CS.AI 50%

arXiv:2602.16512v2 Announce Type: replace Abstract: Prompting schemes such as Chain of Thought, Tree of Thoughts, and Graph of Thoughts can significantly enhance the reasoning capabilities of large language models. However, most existing schemes require users to define static,...

https://arxiv.org/abs/2602.16512

90 CONF

Exploring Plan Space through Conversation: An Agentic Framework for LLM-Mediated Explanations in Planning

ArXiv CS.AI 50%

arXiv:2603.02070v3 Announce Type: replace Abstract: When automating plan generation for a real-world sequential decision problem, the goal is often not to replace the human planner, but to facilitate an iterative reasoning and elicitation process, where the human's role is to...

https://arxiv.org/abs/2603.02070

91 CONF

Chronos: The AI Co-Historian

ArXiv CS.AI 50%

arXiv:2604.03553v2 Announce Type: replace Abstract: AI is increasingly supporting, accelerating, and automating scientific discovery across subjects. Yet, the adoption of AI in historical research remains limited due to the lack of specialised solutions for historians. To change...

https://arxiv.org/abs/2604.03553

92 CONF

TRACE: Capability-Targeted Agentic Training

ArXiv CS.AI 50%

arXiv:2604.05336v2 Announce Type: replace Abstract: Models often fail to complete agentic tasks because they lack core capabilities required by the target environment. However, mainstream approaches for addressing these failures typically either fine-tune directly on target...

https://arxiv.org/abs/2604.05336

93 CONF

HaorFloodAlert: A 72-Hour Machine Learning Early Warning System for Flash Floods in Bangladesh's Haor Wetlands

ArXiv CS.AI 50%

arXiv:2605.20167v2 Announce Type: replace Abstract: Every spring, flash floods strike the haor wetlands of northeast Bangladesh just before the boro rice harvest, and one flood can erase a family's entire crop in days. Warning people in time is hard here for a structural reason:...

https://arxiv.org/abs/2605.20167

94 CONF

Some hypotheses on how chatbots work in problem-solving-driven conversations. Large Language Models as confirmation of the Innovation Illusion

ArXiv CS.AI 50%

arXiv:2606.07722v2 Announce Type: replace Abstract: We discuss the nature of chatbots as conversation partners when discussing the solution of problems. What can chatbots do and what can't they do? We develop hypotheses on how this can this be explained. Our argument draws on...

https://arxiv.org/abs/2606.07722

95 CONF

LLM-Assisted Semantic Alignment and Integration in Collaborative Model-Based Systems Engineering Using SysML v2

ArXiv CS.AI 50%

arXiv:2508.16181v2 Announce Type: replace-cross Abstract: Cross-organizational collaboration in Model-Based Systems Engineering (MBSE) faces many challenges in achieving semantic alignment across independently developed system models. SysML v2 introduces enhanced structural...

https://arxiv.org/abs/2508.16181

96 CONF

Kairos: A Regret-Aware Native World-Action Model Stack for Physical AI

ArXiv CS.AI 50%

arXiv:2606.16533v3 Announce Type: replace Abstract: We introduce \textbf{Kairos}, a regret-aware native world-action model stack for Physical AI. Kairos is motivated by the view that a physical world model should not aim to fully simulate all future pixels, but should learn and...

https://arxiv.org/abs/2606.16533

97 **CONF**

Skill Coverage: A Test Adequacy Metric for Agent Skills

ArXiv CS.AI 50%

arXiv:2606.20659v2 Announce Type: replace Abstract: Agent skills encode reusable procedural knowledge for large language model (LLM) agents, and existing benchmarks show that such skills can improve task-level performance. However, a task outcome does not reveal which parts of a...

https://arxiv.org/abs/2606.20659

98 **CONF**

Querying and Repairing Inconsistent Prioritized Knowledge Bases: Complexity Analysis and Links with Abstract Argumentation

ArXiv CS.AI 50%

arXiv:2003.05746v5 Announce Type: replace-cross Abstract: In this paper, we explore the issue of inconsistency handling over prioritized knowledge bases (KBs), which consist of an ontology, a set of facts, and a priority relation between conflicting facts. In the database...

https://arxiv.org/abs/2003.05746

99 **CONF**

Domain Knowledge-Informed Self-Supervised Representations for Workout Form Assessment

ArXiv CS.AI 50%

arXiv:2202.14019v3 Announce Type: replace-cross Abstract: Maintaining proper form while exercising is important for preventing injuries and maximizing muscle mass gains. Detecting errors in workout form naturally requires estimating human's body pose. However, off-the-shelf pose...

https://arxiv.org/abs/2202.14019

00 **CONF**

Restricted Bernoulli Matrix Factorization: Balancing the trade-off between prediction accuracy and coverage in classification based collaborative filtering

ArXiv CS.AI 50%

arXiv:2210.10619v3 Announce Type: replace-cross Abstract: Reliability measures associated with the prediction of the machine learning models are critical to strengthening user confidence in artificial intelligence. Therefore, those models that provide not only predictions, but...

https://arxiv.org/abs/2210.10619

01 **CONF**

Unveiling the Unborn: Advancing Fetal Health Classification through Machine Learning

ArXiv CS.AI 50%

arXiv:2310.00505v3 Announce Type: replace-cross Abstract: Fetal health classification is a critical task in obstetrics, enabling early identification and management of potential health problems. However, it remains challenging due to data complexity and limited labeled samples....

https://arxiv.org/abs/2310.00505

02 **CONF**

CausalChaos! Dataset for Comprehensive Causal Action Question Answering Over Longer Causal Chains Grounded in Dynamic Visual Scenes

ArXiv CS.AI 50%

arXiv:2404.01299v3 Announce Type: replace-cross Abstract: Causal video question answering (QA) has garnered increasing interest, yet existing datasets often lack depth in causal reasoning. To address this gap, we capitalize on the unique properties of cartoons and construct...

https://arxiv.org/abs/2404.01299

03 **CONF**

Saving GPU Hours in LLM Inference System Development and Online Workloads with Simulation and DBMS-Inspired Cache Replacement Policies

ArXiv CS.AI 50%

arXiv:2411.07447v5 Announce Type: replace-cross Abstract: LLMs are increasingly used world-wide from daily tasks to agentic systems and data analytics, requiring significant GPU resources. While LLM inference systems are capable of serving millions of requests from multiple...

https://arxiv.org/abs/2411.07447

04 **CONF**

Code Benchmarks Should Prioritize Rigor, Reliability, and Reproducibility

ArXiv CS.AI 50%

arXiv:2501.10711v5 Announce Type: replace-cross Abstract: Code-related benchmarks play a critical role in evaluating large language models (LLMs), yet their quality fundamentally shapes how the community interprets model capabilities. In the past few years, awareness of...

https://arxiv.org/abs/2501.10711

05 **CONF**

Evolutionary Guided Decoding: Iterative Value Refinement for LLMs

ArXiv CS.AI 50%

arXiv:2503.02368v4 Announce Type: replace-cross Abstract: While guided decoding, especially value-guided methods, has emerged as a cost-effective alternative for controlling language model outputs without re-training models, its effectiveness is limited by the accuracy of the...

https://arxiv.org/abs/2503.02368

06 **CONF**

Prima.cpp: Fast 30-70B LLM Inference on Heterogeneous and Low-Resource Home Clusters

ArXiv CS.AI 50%

arXiv:2504.08791v3 Announce Type: replace-cross Abstract: On-device inference offers privacy, offline use, and instant response, but consumer hardware restricts large language models (LLMs) to low throughput and capability. To overcome this challenge, we present prima.cpp, a...

https://arxiv.org/abs/2504.08791

07 **CONF**

MOSAIC: Skill-Centric Manipulation Planning with Physics Simulation

ArXiv CS.AI 50%

arXiv:2504.16738v3 Announce Type: replace-cross Abstract: Planning long-horizon manipulation motions using a set of predefined skills is a central challenge in robotics; solving it efficiently could enable general-purpose robots to tackle novel tasks by flexibly composing...

https://arxiv.org/abs/2504.16738

08 **CONF**

Seven Security Challenges in Cross-domain Multi-agent LLM Systems

ArXiv CS.AI 50%

arXiv:2505.23847v5 Announce Type: replace-cross Abstract: Large language models (LLMs) are rapidly evolving into autonomous agents that cooperate across organizational boundaries, enabling joint disaster response, supply-chain optimization, and other tasks that demand...

https://arxiv.org/abs/2505.23847

09 CONF

Leveraging Natural Language Processing to Unravel the Mystery of Life: A Review of NLP Approaches in Genomics, Transcriptomics, and Proteomics

ArXiv CS.AI 50%

arXiv:2506.02212v2 Announce Type: replace-cross Abstract: Natural Language Processing (NLP) has transformed various fields beyond linguistics by applying techniques originally developed for human language to the analysis of biological sequences. This review explores the...

https://arxiv.org/abs/2506.02212

10 CONF

Interaction Techniques that Encourage Longer Prompts Can Improve Psychological Ownership when Writing with AI

ArXiv CS.AI 50%

arXiv:2507.03670v2 Announce Type: replace-cross Abstract: Writing longer prompts for an AI assistant to generate a story increases psychological ownership, a user's feeling that the writing belongs to them. To encourage users to write longer prompts, we evaluated two interaction...

https://arxiv.org/abs/2507.03670

11 CONF

Towards Mitigation of Hallucination for LLM-empowered Agents: Progressive Generalization Bound Exploration and Watchdog Monitor

ArXiv CS.AI 50%

arXiv:2507.15903v2 Announce Type: replace-cross Abstract: Empowered by large language models (LLMs), intelligent agents have become a popular paradigm for interacting with open environments to facilitate AI deployment. However, hallucinations generated by LLMs-where outputs are...

https://arxiv.org/abs/2507.15903

12 CONF

Parity-Aware Byte-Pair Encoding: Improving Cross-lingual Fairness in Tokenization

ArXiv CS.AI 50%

arXiv:2508.04796v3 Announce Type: replace-cross Abstract: Tokenization is the first -- and often least scrutinized -- step of most NLP pipelines. Standard algorithms for learning tokenizers rely on frequency-based objectives, which favor languages dominant in the training data...

https://arxiv.org/abs/2508.04796

13 CONF

Context Misleads LLMs: The Role of Context Filtering in Maintaining Safe Alignment of LLMs

ArXiv CS.AI 50%

arXiv:2508.10031v2 Announce Type: replace-cross Abstract: While Large Language Models (LLMs) have shown significant advancements in performance, various jailbreak attacks have posed growing safety and ethical risks. Malicious users often exploit adversarial context to deceive...

https://arxiv.org/abs/2508.10031

14 CONF

ChainReaction: Causal Chain-Guided Reasoning for Modular and Explainable Causal-Why Video Question Answering

ArXiv CS.AI 50%

arXiv:2508.21010v3 Announce Type: replace-cross Abstract: Existing Causal-Why Video Question Answering (VideoQA) models often struggle with higher-order reasoning, relying on opaque, monolithic pipelines that entangle video understanding, causal inference, and answer generation....

https://arxiv.org/abs/2508.21010

15 **CONF**

Agentic Artificial Intelligence for Multistage Physics Experiments at a Large-Scale User Facility Particle Accelerator

ArXiv CS.AI **50%**

arXiv:2509.17255v2 Announce Type: replace-cross Abstract: We present the first language-model-driven agentic artificial intelligence (AI) system to autonomously execute multi-stage physics experiments on a production synchrotron light source. Implemented at the Advanced Light...

https://arxiv.org/abs/2509.17255

16 **CONF**

Polarity Detection of Sustainable Development Goals in News Text

ArXiv CS.AI **50%**

arXiv:2509.19833v4 Announce Type: replace-cross Abstract: The United Nations' Sustainable Development Goals (SDGs) provide a globally recognised framework for addressing major societal, environmental, and economic challenges. While recent advances in natural language processing...

https://arxiv.org/abs/2509.19833

17 **CONF**

MAD-PINN: A Decentralized Physics-Informed Machine Learning Framework for Safe and Optimal Multi-Agent Control

ArXiv CS.AI **50%**

arXiv:2509.23960v2 Announce Type: replace-cross Abstract: Co-optimizing safety and performance in large-scale multi-agent systems remains a fundamental challenge. Existing approaches based on multi-agent reinforcement learning (MARL), safety filtering, or Model Predictive...

https://arxiv.org/abs/2509.23960

18 **CONF**

SoK: Systematizing LLM Prompt Security: Taxonomies, Datasets, and Unified Evaluation of Attacks and Defenses

ArXiv CS.AI **50%**

arXiv:2510.15476v3 Announce Type: replace-cross Abstract: Large Language Models (LLMs) are increasingly used as interfaces to information, code, and real-world services, making prompt-level security failures a practical concern. Although jailbreak attacks, defenses, datasets,...

https://arxiv.org/abs/2510.15476

19 **CONF**

PuzzleMoE: Efficient Compression of Large Mixture-of-Experts Models via Sparse Expert Merging and Bit-packed inference

ArXiv CS.AI **50%**

arXiv:2511.04805v2 Announce Type: replace-cross Abstract: Mixture-of-Experts (MoE) models have shown strong potential in scaling language models efficiently by activating only a small subset of experts per input. However, their widespread deployment remains limited due to the...

https://arxiv.org/abs/2511.04805

20 **CONF**

InverseCrafter: Efficient Video ReCapture as a Latent Domain Inverse Problem

ArXiv CS.AI **50%**

arXiv:2512.05672v2 Announce Type: replace-cross Abstract: Recent approaches in controllable novel view video generation often rely on fine-tuning pre-trained Video Diffusion Models (VDMs). This dominant paradigm is computationally expensive and frequently suffers from...

https://arxiv.org/abs/2512.05672

21 **CONF**

Developing an LLM-Based Feedback System Grounded in Evidence-Centered Design to Support Physics Problem Solving

ArXiv CS.AI 50%

arXiv:2512.10785v3 Announce Type: replace-cross Abstract: Generative AI offers new opportunities for individualized and adaptive learning, e.g., through large language model (LLM)-based feedback systems. While LLMs can produce factually correct feedback for relatively...

https://arxiv.org/abs/2512.10785

22 **CONF**

EvoXplain: When Machine Learning Models Agree on Predictions but Disagree on Why -- Measuring Mechanistic Multiplicity Across Training Runs

ArXiv CS.AI 50%

arXiv:2512.22240v5 Announce Type: replace-cross Abstract: Machine learning models are primarily judged by predictive performance, especially in applied genomics, where explanations are read as biological findings. In practice, reported gene panels are stabilised by averaging,...

https://arxiv.org/abs/2512.22240

23 **CONF**

JADAI: Jointly Amortizing Adaptive Design and Bayesian Inference

ArXiv CS.AI 50%

arXiv:2512.22999v2 Announce Type: replace-cross Abstract: We consider problems of parameter estimation where design variables can be actively optimized to maximize information gain. To this end, we introduce JADAI, a framework that jointly amortizes Bayesian adaptive design and...

https://arxiv.org/abs/2512.22999

24 **CONF**

Resource-constrained Project Scheduling with Time-of-Use Energy Tariffs and Machine States: A Logic-based Benders Decomposition Approach

ArXiv CS.AI 50%

arXiv:2601.06542v2 Announce Type: replace-cross Abstract: In this paper, we investigate the Resource-Constrained Project Scheduling Problem (RCPSP) with Time-of-Use (TOU) energy tariffs and machine states, a variant of RCPSP for production scheduling, where energy price is part...

https://arxiv.org/abs/2601.06542

25 **CONF**

ARCQuant: Boosting NVFP4 Quantization with Augmented Residual Channels for LLMs

ArXiv CS.AI 50%

arXiv:2601.07475v2 Announce Type: replace-cross Abstract: The emergence of fine-grained numerical formats like NVFP4 presents new opportunities for efficient Large Language Model (LLM) inference. However, it is difficult to adapt existing Post-Training Quantization (PTQ)...

https://arxiv.org/abs/2601.07475

26 **CONF**

FlashBlock: Attention Caching for Efficient Long-Context Block Diffusion

ArXiv CS.AI 50%

arXiv:2602.05305v3 Announce Type: replace-cross Abstract: Generating long-form content, such as minute-long videos and extended texts, is increasingly important for modern generative models. Block diffusion improves inference efficiency via KV caching and block-wise causal...

https://arxiv.org/abs/2602.05305

27 CONF

Multi-Way Representation Alignment

ArXiv CS.AI 50%

arXiv:2602.06205v2 Announce Type: replace-cross Abstract: The Platonic Representation Hypothesis suggests that independently trained neural networks converge to increasingly similar latent spaces. However, current strategies for mapping these representations are inherently...

https://arxiv.org/abs/2602.06205

28 CONF

pFedNavi: Structure-Aware Personalized Federated Vision-Language Navigation for Embodied AI

ArXiv CS.AI 50%

arXiv:2602.14401v2 Announce Type: replace-cross Abstract: Vision-Language Navigation VLN requires large-scale trajectory instruction data from private indoor environments, raising significant privacy concerns. Federated Learning FL mitigates this by keeping data on-device, but...

https://arxiv.org/abs/2602.14401

29 CONF

Gradient Regularization Mitigates Reward Hacking in Reinforcement Learning from Human Feedback and Verifiable Rewards

ArXiv CS.AI 50%

arXiv:2602.18037v2 Announce Type: replace-cross Abstract: Reinforcement Learning from Human Feedback (RLHF) or Verifiable Rewards (RLVR) are two key steps in the post-training of modern Language Models (LMs). A common problem is reward hacking, where the policy may exploit...

https://arxiv.org/abs/2602.18037

30 CONF

Causal Mechanism Reduction: Mechanism Replacement for Neural Network Pruning and Abstraction

ArXiv CS.AI 50%

arXiv:2602.24266v2 Announce Type: replace-cross Abstract: Which internal mechanisms of a neural network can be replaced while preserving the computation it performs? Structured pruning asks for smaller deployable networks; causal abstraction asks for high-level models that...

https://arxiv.org/abs/2602.24266

31 CONF

BEVLM: Distilling Semantic Knowledge from LLMs into Bird's-Eye View Representations

ArXiv CS.AI 50%

arXiv:2603.06576v2 Announce Type: replace-cross Abstract: The integration of Large Language Models (LLMs) into autonomous driving has attracted growing interest for their strong reasoning and semantic understanding abilities, which are essential for handling complex...

https://arxiv.org/abs/2603.06576

32 CONF

A Hybrid Quantum Circuit Born Machine Framework for Financial Volatility Forecasting: Quantum-Assisted Training and Classical Inference

ArXiv CS.AI 50%

arXiv:2603.09789v3 Announce Type: replace-cross Abstract: Accurate financial volatility forecasting is crucial but challenged by the non-linear, highly correlated nature of market data. Recently, quantum computing has emerged as a promising paradigm for solving complex...

https://arxiv.org/abs/2603.09789

33 CONF

Reachability Across the NL/PL Boundary: A Taxonomy-Driven Dataflow Model for LLM-Integrated Applications

ArXiv CS.AI 50%

arXiv:2603.28345v3 Announce Type: replace-cross Abstract: LLM API calls have become a standard programming primitive, but they create a program boundary that disrupts traditional dataflow analysis. A runtime value may be inserted into a natural-language prompt through a template...

https://arxiv.org/abs/2603.28345

34 CONF

Don't Make Models Guess Security and Safety: Symbolic Guardrails for Domain-Specific AI Agents

ArXiv CS.AI 50%

arXiv:2604.15579v2 Announce Type: replace-cross Abstract: There is increasing interest in integrating AI agents that invoke tools into domain-specific commercial software, where unintended tool calls can cause serious security and safety incidents. This has drawn growing...

https://arxiv.org/abs/2604.15579

35 CONF

Governed MCP: Kernel-Level Tool Governance for AI Agents via Logit-Based Safety Primitives

ArXiv CS.AI 50%

arXiv:2604.16870v2 Announce Type: replace-cross Abstract: AI agents increasingly call external tools (file system, network, APIs) through the Model Context Protocol (MCP). These tool calls are the agent's syscalls: privileged operations with side effects on shared state, yet...

https://arxiv.org/abs/2604.16870

36 CONF

Test-Time Adaptation for EEG Foundation Models: A Systematic Study under Real-World Distribution Shifts

ArXiv CS.AI 50%

arXiv:2604.16926v2 Announce Type: replace-cross Abstract: Electroencephalography (EEG) foundation models have shown strong potential for learning generalizable representations from large-scale neural data, yet their clinical deployment is hindered by distribution shifts across...

https://arxiv.org/abs/2604.16926

37 CONF

Towards Generalizable Deepfake Image Detection with Vision Transformers

ArXiv CS.AI 50%

arXiv:2604.17376v2 Announce Type: replace-cross Abstract: In today's day and age, we face a challenge in detecting deepfake images because of the fast evolution of modern generative models and the poor generalization capability of existing methods. In this paper, we use an...

https://arxiv.org/abs/2604.17376

38 CONF

CorridorVLA: Explicit Spatial Constraints for Generative Action Heads via Sparse Anchors

ArXiv CS.AI 50%

arXiv:2604.21241v2 Announce Type: replace-cross Abstract: Vision--Language--Action (VLA) models often use intermediate representations to connect multimodal inputs with continuous control, yet spatial guidance is often injected implicitly through latent features. We propose...

https://arxiv.org/abs/2604.21241

39 CONF

Incompressible Knowledge Probes: Estimating Black-Box LLM Parameter Counts via Factual Capacity

ArXiv CS.AI 50%

arXiv:2604.24827v2 Announce Type: replace-cross Abstract: Closed-source frontier labs do not disclose parameter counts. Storing F facts requires at least F/(bits per parameter) weights, so factual recall lower-bounds parameter count--an intrinsic, serving-independent signal,...

https://arxiv.org/abs/2604.24827

40 CONF

Clin-JEPA: A Multi-Phase Co-Training Framework for Joint-Embedding Predictive Pretraining on EHR Patient Trajectories

ArXiv CS.AI 50%

arXiv:2605.10840v4 Announce Type: replace-cross Abstract: We present Clin-JEPA, a multi-phase co-training framework for joint-embedding predictive (JEPA) pretraining on EHR patient trajectories. JEPA architectures have enabled latent-space planning in robotics and high-quality...

https://arxiv.org/abs/2605.10840

41 CONF

High-Risk AI Systems and the Problem of Identity in the European AI Act

ArXiv CS.AI 50%

arXiv:2605.23922v3 Announce Type: replace-cross Abstract: The EU Artificial Intelligence Act (AIA) establishes a lifecycle governance regime for high-risk AI systems built around ex-ante conformity assessment, post-market monitoring, and re-assessment upon "substantial...

https://arxiv.org/abs/2605.23922

42 CONF

De-attribute to Forget for LLM Unlearning

ArXiv CS.AI 50%

arXiv:2605.30919v2 Announce Type: replace-cross Abstract: The rapid development of large language models (LLMs) has raised concerns on the use of inappropriate data for training, which has led to a growing interest in LLM unlearning. Many existing LLM unlearning approaches rely...

https://arxiv.org/abs/2605.30919

43 CONF

Automatically Differentiable Nonlinear Tensor Networks (ADNTNs) for Exponential Parameter Compression of Deep Neural Networks

ArXiv CS.AI 50%

arXiv:2606.00130v2 Announce Type: replace-cross Abstract: Large deep neural networks are costly to store and deploy because inference must move and evaluate many parameters. This paper studies \emph{Automatically Differentiable Nonlinear Tensor Networks} (ADNTNs), compact...

https://arxiv.org/abs/2606.00130

44 CONF

A Mathematical Theory of Value: a synthesis on goal-directed agency under resource constraints

ArXiv CS.AI 50%

arXiv:2606.12502v2 Announce Type: replace-cross Abstract: We propose that value -- the quantity goal-directed agents create, destroy, and exchange -- is a lawful structural quantity in the same category as information. Following Shannon's method, we make one ruthless...

https://arxiv.org/abs/2606.12502

45 **CONF**

Attention is Just Another Name for Coupling? A Fast-Slow ODE Perspective on Hierarchical Pretraining

ArXiv CS.AI 50%

arXiv:2606.16730v2 Announce Type: replace-cross Abstract: We re-interpret Transformer pretraining as a fast-slow, singularly perturbed flow along depth, with untied weights as its non-autonomous feature. The linearised dynamics is a depth-ordered product of layer maps. Along a...

https://arxiv.org/abs/2606.16730

46 **CONF**

The Power of Light: Improving Synthetic-to-Real Domain Adaptation through Physically-Based Indirect Illumination

ArXiv CS.AI 50%

arXiv:2606.22574v2 Announce Type: replace-cross Abstract: While synthetic data generation resolves the manual labeling bottleneck in computer vision, minimizing the syn-to-real domain gap requires optimizing rendering variables. This paper presents a systematic study analyzing...

https://arxiv.org/abs/2606.22574

47 **CONF**

UC-Search: Risk-Aware Test-Time Search for Delayed Constrained Time-Series Control

ArXiv CS.AI 50%

arXiv:2606.25274v2 Announce Type: replace-cross Abstract: Time-series deployments often need delayed feasible decisions, not only accurate forecasts. UC-Search is a trace-only retained-search layer for delayed constrained control: a frozen backbone emits forecasts or action...

https://arxiv.org/abs/2606.25274

48 **CONF**

The Unverifiability of Artificial General Intelligence (AGI) Alignment, Static and Dynamic: From Trakhtenbrot's Wall to the Safety-Generality Tension

ArXiv CS.AI 50%

arXiv:2606.28639v2 Announce Type: replace-cross Abstract: We establish the mathematical limits of AGI safety in two forms: verifying a fixed system, and verifying that a certified safety property persists once the system self-modifies. In the static case, no algorithm can...

https://arxiv.org/abs/2606.28639

49 **CONF**

Reward-Free Code Alignment from Pretrained or Fine-Tuned LLM: Unpacking the Trade-offs for Code Generation

ArXiv CS.AI 50%

arXiv:2606.28998v2 Announce Type: replace-cross Abstract: Large Language Model (LLM) alignment trains an LLM using preference data to produce outputs that better meet established quality standards. While LLM alignment techniques are studied for non-coding tasks, we know little...

https://arxiv.org/abs/2606.28998

50 **CONF**

From Materials Database to Materials Bank: Assetizing Data for AI Driven Materials Innovation

ArXiv CS.AI 50%

arXiv:2606.31366v2 Announce Type: replace-cross Abstract: Driven by high-throughput experimentation, computational modeling, and artificial intelligence (AI), materials data has expanded at an unprecedented rate. Conventional materials databases function only as passive...

https://arxiv.org/abs/2606.31366